

Government Reputation, FDI, and Profit Shifting*

Yeonggyu Yun[†]

September 2026

Abstract

I document that countries with higher investment risk have higher statutory corporate tax rates. I rationalize this fact in a model where investment risk arises from multinational firms' uncertainty about the government's commitment to announced tax rates. A commitment type must honor its announcement, whereas an opportunistic type can revise taxes after investment. Reputation is modeled as the probability firms assign to the commitment type. Profit shifting allows firms to retain their returns even after a tax revision. In the static model, a marginal increase in the announced tax reduces expected returns proportionally more when reputation is higher. The tax base becomes more elastic, inducing the commitment type to lower its tax. Without profit shifting, the tax base becomes less elastic as reputation rises, and the optimal tax increases. In the dynamic model, a tax revision reveals the opportunistic type and reduces future investment and revenue. Higher reputation increases the future revenue cost of a tax revision, supporting lower sustainable taxes and greater investment. Incorporating profit shifting in the dynamic model explains why investment and production can persist under low reputation, whereas investment vanishes as in the model without profit shifting.

Keywords: Commitment, Corporate taxation, Foreign direct investment, Government reputation, Multinational firm, Profit shifting

JEL Classification: H21, H25, H26, F21, F23

*I am deeply indebted to Charles Engel, Rishabh Kirpalani, and Kim Ruhl for advice and guidance. I appreciate comments from Carter Braxton, Louphou Coulibaly, Lydia Cox, Agustín Gutiérrez, Nora Strecker, Heejin Yoon, and seminar participants at the University of Wisconsin-Madison, the Fall 2023 Midwest Macro Meeting, the 2024 International Institute of Public Finance Congress, and the Summer 2024 Asian Meeting of the Econometric Society. Financial support from the University of Wisconsin-Madison and the Korea Foundation for Advanced Studies is gratefully acknowledged.

[†]University of Wisconsin-Madison. Email: yyun23@wisc.edu.

1 Introduction

A country with greater investment risk might offer a lower corporate tax rate to attract foreign investment. I document the opposite relationship. Countries with higher investment risk have higher statutory corporate tax rates. I study how multinational firms' ability to shift profits changes the taxes that governments choose and the investment they can sustain when firms are uncertain about government commitment.

I measure investment risk using the Investment Profile of the International Country Risk Guide (ICRG). The index covers contract viability, expropriation, profit repatriation, and payment delays. In a panel of 101 countries, higher risk is associated with higher statutory corporate tax rates after controlling for economic, institutional, and political conditions. I also examine foreign-affiliate investment around the 2018 *Achmea* judgment, which restricted investor–state arbitration under intra-EU treaties. Investment declined in treaty-covered country pairs relative to uncovered pairs.

The model focuses on investment risk arising from uncertainty about the government's commitment to future tax policy. Multinational firms allocate capital between a host country and a foreign location. The host government maximizes corporate tax revenue. A commitment type must implement the tax announced before investment, whereas an opportunistic type can change that rate after firms invest. I refer to reneging on a tax promise after investment as a *tax revision*. Reputation is the probability that firms assign to the commitment type. I study equilibria in which both types announce the same tax so firms remain uncertain about government type when investing. After observing the implemented tax, firms can shift reported profits abroad while leaving physical capital in place.

In the one-period model with profit shifting, higher reputation makes the tax base more elastic with respect to the announced tax rate. Firms retain returns even when the government revises the announced tax, so a marginal increase in the announced rate reduces expected returns by a larger proportion when reputation is higher. The larger proportional contraction of the tax base induces the commitment type to lower its rate. Without profit shifting, firms retain host profits only under the commitment type. Higher reputation then makes the tax base less elastic, and the optimal announced tax rises. Profit shifting thus reverses the relationship between reputation and the tax under commitment, rationalizing the observed association between investment risk and statutory rates.

I then extend the model to two periods. The same firms invest in each period, and capital fully depreciates after production. Firms observe past tax policy before investing again, so a tax revision reveals the opportunistic type and reduces future investment and revenue. The loss of future revenue can induce the opportunistic type to honor its

announcement. In a numerical example with profit shifting, the optimal first-period tax falls with reputation until it reaches the full-commitment rate. Without profit shifting, the optimal tax first rises and then falls. At higher reputation, the larger future revenue loss from a tax revision allows the government to sustain a lower announced tax.

I characterize sustainable tax policies and investment plans by requiring government and firm decisions to be sequentially optimal. Among the taxes that both types of government implement, higher reputation allows for a lower tax rate and encourages investment. The opportunistic type loses more future revenue by revising the announced tax when initial reputation is higher. This loss supports tax reductions until the full-commitment rate becomes sustainable.

Profit shifting also changes how much investment can be sustained at low reputation. Over any fixed finite horizon, taxes implemented by both types before the final period approach the discretionary rate as the probability of the commitment type vanishes. Without profit shifting, investment and production vanish. With profit shifting, firms retain part of their returns under discretionary taxation, and investment remains positive. The investment comparison continues to hold when tax announcements and implementation depend on policy history. Profit shifting therefore supports real activity even when reputation can no longer sustain a tax below the government's preferred rate on installed capital.

Literature Review This paper contributes to three strands of literature. First, I contribute to research on capital taxation and investment under limited commitment. This literature studies how the prospect of future investment restrains a government's incentive to tax installed capital ([Chari et al., 1988](#); [Thomas and Worrall, 1994](#); [Aguiar and Amador, 2011](#); [Schnitzer, 1999](#)). [Chari and Kehoe \(1990\)](#) formulate sustainable plans, and [Phelan and Stacchetti \(2001\)](#) characterize sequential equilibria in a Ramsey tax model. I show how profit shifting changes the investment consequences of weak commitment. Firms can retain returns on installed capital under discretionary taxation, so positive investment can persist even when future revenue no longer induces the government to keep taxes below its discretionary rate.

Second, I contribute to the literature on government reputation and policy credibility. [Backus and Driffill \(1985\)](#) and [King et al. \(2008\)](#) study how learning about policymakers shapes monetary policy and the credibility of announcements. [Phelan \(2006\)](#) links tax policy and public trust when the government type can change over time, while [Dovis and Kirpalani \(2021\)](#) study policy rules under uncertainty about commitment. I show that reputation affects investment through the tax rate that the government can sustain, even when both types already honor their announcements with certainty. Higher reputation increases the future revenue loss after a tax revision and allows for a

lower rate to satisfy the implementation incentive compatibility constraint. Investment rises through this reduction in the sustainable tax. The probability of honoring an announcement alone, therefore, does not determine the investment cost of limited commitment. An announcement may be certain to be implemented precisely because the tax is high enough to discourage a tax revision.

Third, I contribute to the literature on profit shifting and corporate taxation. [Hong and Smart \(2010\)](#) show how international tax planning allows governments to tax corporate income while retaining mobile investment. Other studies examine how profit shifting changes tax competition ([Peralta et al., 2006](#); [Devereux et al., 2008](#); [Johannesen, 2010](#); [Mongrain et al., 2023](#)). Closest to this paper, [Mardan and Stimmelmayer \(2020\)](#) show that profit shifting can reverse the relationship between country risk and corporate tax rates when risk is an exogenous probability of project failure. I explain why a similar relationship arises when the government can change the tax treatment of investment. Profit shifting preserves returns under discretionary taxation, so a promised tax cut has a smaller proportional effect on expected returns when commitment is less likely. The commitment type then faces a less elastic tax base and chooses a higher rate. The risk–tax relation thus depends on how firms respond to tax promises under uncertain commitment. The dynamic analysis then determines which tax announcements the value of future investment can sustain.

Outline Section 2 presents the empirical evidence. Section 3 develops the model, and Section 4 derives the relation between reputation and optimal taxation in one period. Section 5 introduces future investment and endogenous tax implementation. Section 6 characterizes sustainable tax policies and investment over two periods and a fixed finite horizon. Section 7 concludes.

2 Investment Risk and Corporate Taxation

2.1 Investment Risk and Statutory Tax Rates

I measure investment risk using the ICRG Investment Profile. The index combines Contract Viability/Expropriation, Profits Repatriation, and Payment Delays. Each component has a maximum of four points, with higher scores denoting lower risk ([The PRS Group, nd](#)). The three components concern the security of firms’ contracts and assets and the ability to receive and transfer investment returns.

For country h and year t , I define investment risk as

$$\text{Risk}_{ht} = 12 - \text{Investment Profile}_{ht},$$

TABLE 1: Investment risk and statutory corporate tax rates

	(1)	(2)	(3)	(4)	(5)
Investment risk	0.395	0.394	0.435	0.426	0.420
Standard error	(0.148)	(0.149)	(0.142)	(0.143)	(0.145)
<i>p</i> -value	0.009	0.009	0.003	0.004	0.005
Economic controls	Yes	Yes	Yes	Yes	Yes
Institutional controls	No	Yes	No	Yes	Yes
Political controls	No	No	Yes	Yes	Yes
Lagged GDP growth	No	No	No	No	Yes
Country and year effects	Yes	Yes	Yes	Yes	Yes
Observations	1,887	1,887	1,887	1,887	1,887
Countries	101	101	101	101	101
Within R^2	0.025	0.031	0.033	0.038	0.039

Notes: OECD annual rates over 2003–2021. All columns use the same 1,887 observations in 101 countries and include country and year effects. Economic controls are log real GDP per capita, log population, and resource rents/GDP. Institutional controls are corruption, law and order, and bureaucratic quality. Political controls are government stability, internal conflict, and external conflict. All ICRG controls are lagged risks. Parentheses contain country-clustered standard errors. Appendix Table 3 gives all slope coefficients for columns (4) and (5).

so that higher values indicate greater risk. I average the score across months and use the annual score with a one-year lag in the regressions. The available data report the combined score rather than the three subcomponents.

I use the OECD combined statutory corporate income tax rate, including central and subnational taxation (OECD, 2026). The statutory rate is specified in tax law rather than calculated from firms' payments and profits. The annual observations do not identify whether a rate was announced before a particular investment or revised afterward.

The sample contains 1,887 country-year observations in 101 countries over 2003–2021.¹ I estimate

$$\text{Tax rate}_{ht} = \alpha_h + \lambda_t + \beta \text{Risk}_{h,t-1} + \delta' X_{ht} + \phi' Z_{h,t-1} + u_{ht}. \quad (1)$$

Country effects α_h absorb persistent country differences, and year effects λ_t absorb common changes. The vector X contains log real GDP per capita, log population, and natural-resource rents/GDP. The vector Z contains ICRG measures of corruption, law and order, bureaucratic quality, government stability, internal conflict, and external conflict. These separate ICRG components control for institutional and political conditions that may affect both investment risk and taxation (The PRS Group, nd). All ICRG measures are expressed as risks and lagged one year.

¹I exclude twenty zero-rate observations inconsistent with historical tax descriptions. Appendix A.1 documents the exclusions, and Appendix A.2 reports estimates retaining the original records.

With economic, institutional, and political controls, a one-point increase in investment risk is associated with a 0.426 percentage-point higher statutory tax rate ($p = 0.004$, column (4) of Table 1). The coefficient is positive and statistically significant across the reported specifications, including when lagged real GDP growth is added.

The positive association also holds when I retain the questionable zero-rate observations or exclude Haiti and Togo entirely (Appendix Table 4). The regressions describe conditional associations between investment risk and statutory rates.

2.2 Investment Following a Change in Treaty Protection

The risk–tax association leaves open how foreign investment responds when protection against government actions changes. On March 6, 2018, the Court of Justice of the European Union ruled in *Achmea* that the investor–state arbitration clause in an intra-EU bilateral investment treaty was incompatible with EU law (Court of Justice of the European Union, 2018). The judgment restricted treaty-based arbitration against host governments. I examine whether investment by firms covered by intra-EU treaties changed after the judgment.

The investment data come from Eurostat’s inward foreign affiliates statistics (Eurostat, 2026). The series reports gross investment in tangible goods by foreign-controlled enterprises in millions of euros for each host and ultimate-parent country pair over 2010–2019. I identify treaty coverage just before *Achmea* from the lists attached to the termination agreement (European Union, 2020). For inclusion, both the host and the parent country must be parties to that agreement. I drop pairs whose treaties have already been terminated. There are 22 host countries and 22 parent countries in the resulting panel.

I estimate an event-study specification using Poisson pseudo-maximum likelihood with country-pair, host-year, and parent-year fixed effects. The baseline specification compares investment between covered and uncovered pairs relative to 2017. A second specification allows investment in each country pair to follow a different linear trend. Standard errors are clustered by host and parent country. Appendix A.3 gives the estimating equation and reports annual estimates.

Both specifications in Table 2 show a decline in investment in treaty-covered pairs relative to uncovered pairs after the judgment. Differences in pre-event trends and the decline in reporting coverage in 2018 prevent attributing the decrease solely to the judgment. Appendix A.3 reports the corresponding checks.

Host-year effects absorb annual host-country conditions common to affiliates from different parent countries, including changes in the general statutory tax rate. The estimates therefore describe investment differences associated with treaty coverage

TABLE 2: Investment after the Achmea judgment

	(1)	(2)
Average post-judgment coefficient	−0.474** (0.215)	−0.648** (0.280)
<i>p</i> -value	0.039	0.031
Country-pair linear trends	No	Yes
Pre-period joint <i>p</i> -value	0.017	0.549
Observations	2,622	2,547
Host and parent countries	22 each	22 each
Country pairs	349	320

Notes: Each column reports an annual event-study regression estimated by Poisson pseudo-maximum likelihood over 2010–2019. The post-judgment coefficient averages the 2018 and 2019 coefficients. Both regressions include pair, host-year, and parent-year effects. Column (1) uses 2017 as the reference year. Column (2) also includes a separate linear time trend for each country pair and uses 2010 and 2017 as reference years. Parentheses contain standard errors clustered by host and parent country. The joint pre-period tests use host-clustered covariance matrices. ***, **, and * indicate significance at 1, 5, and 10 percent.

around the judgment. The investment comparison does not identify a response to a change in corporate tax rates.

3 Model

The ICRG Investment Profile concerns the security of firms’ claims and their ability to receive and transfer investment returns (The PRS Group, nd). Contract viability and protection against expropriation depend in part on the constraints that governments face when dealing with investors. These constraints matter after investment because installed capital is exposed to subsequent policy changes. Firms, therefore, need to assess whether the policies announced before investment will remain in force afterward. World Bank (2004, pp. 45–50) emphasizes the importance of policy credibility for investment and relates the Investment Profile to constraints on government discretion.

I model investment risk arising from uncertainty about the government’s commitment to taxation. A commitment type must implement the tax announced before investment, whereas an opportunistic type can revise the tax after firms invest. Firms do not observe government type. Reputation p is the probability firms assign to the commitment type. Lower reputation means that firms regard the tax commitment as less likely to bind the government. Firms base investment on expected after-tax returns under the two government types.

3.1 Environment and Timing

There is a unit measure of identical foreign-owned multinational firms. Each firm has access to a fixed amount of capital $\bar{k} > 0$ and allocates it between the host country and a foreign location. Capital employed in the host produces output zk^α , where $z > 0$ and $0 < \alpha < 1$. The rental cost per unit of capital is $r^* \geq 0$ and is deductible where capital is employed. Operating profits are therefore $\rho(k) = zk^\alpha - r^*k$ in the host and $\rho(\bar{k} - k)$ abroad. The host is a small open economy that takes the foreign corporate tax rate $\tau^* \in (0, 1)$ as given.

Both government types maximize corporate tax revenue. Each type applies a uniform tax rate to all firms. The types differ only in whether they can revise the announced tax after investment.

The government announces a tax τ^R before firms invest and implements a rate $\tau \in [0, 1]$ afterward. Firms then choose a profit transfer θ . A positive transfer moves reported profits into the host, and a negative transfer moves profits abroad. All tax announcements and implemented rates are public. The timing is

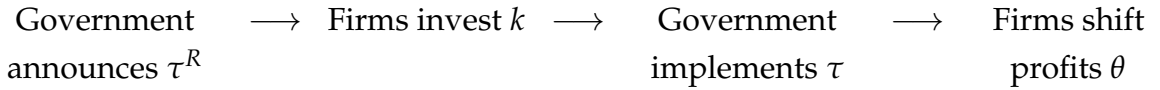


FIGURE 1: Timing within a period

Installed capital is fixed when the government implements the tax. Firms can still adjust intragroup payments before filing taxes. This distinction between investment and reporting is consistent with the within-year reporting adjustments documented by [Hopland et al. \(2018\)](#). Firms anticipate their reporting response when investing. I solve the model by backward induction.

3.2 Firms' Profit-Shifting Choice

Profit shifting incurs a convex resource cost in the country from which profits are transferred. For an outward transfer, the host affiliate incurs $C_H(k, \theta) = \gamma\theta^2/[2\rho(k)]$, and the foreign affiliate incurs no cost, i.e., $C_F(k, \theta) = 0$. For an inward transfer, $C_H(k, \theta) = 0$ and $C_F(k, \theta) = \gamma\theta^2/[2\rho(\bar{k} - k)]$. Both costs are zero when $\theta = 0$. The parameter $\gamma > 0$ measures the cost of shifting profits. Each affiliate deducts these costs from its taxable profits, and each country's tax base must be nonnegative. An affiliate with no operating profits cannot transfer profits to the other country.

I assume that production and profit-shifting costs satisfy

$$\begin{aligned}
& \alpha z \bar{k}^{\alpha-1} > (2 - \alpha)r^*, \\
& \gamma > \frac{\tau^*}{1 - \tau^*} + \frac{(\tau^*)^2}{2(1 - \tau^*)^2}, \\
& \frac{\tau^*}{\gamma(1 - \tau^*)} \left(\frac{\sqrt{1 + 2\gamma} + 1}{2} \right)^{\alpha/(1-\alpha)} + \frac{\tau^* \alpha}{(1 - \alpha)(1 - \tau^*)} < 1, \\
& \frac{\sqrt{1 + 2\gamma} - 1}{2} \left[\frac{1}{2} + \frac{(2 - \alpha)r^*}{\alpha z (\bar{k}/2)^{\alpha-1}} \right] < 1 - \alpha.
\end{aligned} \tag{2}$$

The production restriction ensures positive marginal profits on the investment domain. Firms may transfer profits in either direction.

Given the capital and the implemented tax, the firm solves

$$\begin{aligned}
& \max_{\theta} \quad (1 - \tau)[\rho(k) + \theta - C_H(k, \theta)] \\
& \quad + (1 - \tau^*)[\rho(\bar{k} - k) - \theta - C_F(k, \theta)], \\
& \text{subject to} \quad \rho(k) + \theta - C_H(k, \theta) \geq 0, \\
& \quad \rho(\bar{k} - k) - \theta - C_F(k, \theta) \geq 0.
\end{aligned} \tag{3}$$

Let $\bar{y} = 2/(\sqrt{1 + 2\gamma} + 1)$ denote the largest share of operating profits that can be transferred without making the transferring affiliate's tax base negative. The optimal transfer is

$$\theta(k, \tau) = \begin{cases} -\rho(k) \min \left\{ \frac{\tau - \tau^*}{\gamma(1 - \tau)}, \bar{y} \right\}, & \tau^* \leq \tau < 1, \\ \rho(\bar{k} - k) \min \left\{ \frac{\tau^* - \tau}{\gamma(1 - \tau^*)}, \bar{y} \right\}, & \tau < \tau^*. \end{cases} \tag{4}$$

At $\tau = 1$, the outward transfer share equals $\bar{y}$. A tax above the foreign rate induces outward profit shifting, while a tax below the foreign rate induces inward profit shifting.

For outward transfers, write $y(\tau) = -\theta(k, \tau)/\rho(k)$. The after-tax return from one unit of host profit and the government's revenue from that profit are

$$\begin{aligned}
A(\tau) &= (1 - \tau) \left[1 - y(\tau) - \frac{\gamma y(\tau)^2}{2} \right] + (1 - \tau^*)y(\tau), \\
v(\tau) &= \tau \left[1 - y(\tau) - \frac{\gamma y(\tau)^2}{2} \right].
\end{aligned} \tag{5}$$

The firm retains $A(\tau)\rho(k) + (1 - \tau^*)\rho(\bar{k} - k)$ after taxes and reporting costs. At a tax rate of one, $A(1) = (1 - \tau^*)\bar{y} > 0$ because the firm can report profits abroad.

3.3 Taxation after Investment

With a unit measure of identical firms, aggregate host capital in a symmetric allocation is $K = k$. Government revenue at tax τ is

$$\mathcal{R}(K, \tau) = \tau[\rho(K) + \theta(K, \tau) - C_H(K, \theta(K, \tau))].$$

The commitment type implements τ^R . The opportunistic type chooses

$$\tau^O(K) \in \arg \max_{\tau \in [0,1]} \mathcal{R}(K, \tau). \quad (6)$$

The opportunistic type takes installed capital as given but anticipates firms' reporting responses. A higher tax raises revenue per unit of reported profit and reduces the reported tax base through profit shifting.

3.4 Firms' Investment Choice

Firms invest after observing the announcement and before observing the implemented tax. When both government types announce the same rate, the announcement conveys no information about the type. Firms expect τ^R with probability p and $\tau^O(K)$ with probability $1 - p$. Each firm chooses

$$\begin{aligned} \max_{k \in [0, \bar{k}]} \mathbb{E}_\tau \Big\{ & (1 - \tau)[\rho(k) + \theta(k, \tau) - C_H(k, \theta(k, \tau))] \\ & + (1 - \tau^*)[\rho(\bar{k} - k) - \theta(k, \tau) - C_F(k, \theta(k, \tau))] \Big\}. \end{aligned} \quad (7)$$

An atomistic firm takes the implemented taxes as given. It does not internalize the effect of aggregate investment on the government's tax choice. Investment and discretionary taxation must jointly satisfy individual optimality, $k = K$, and (6). I denote the resulting host investment by $K(\tau^R, p)$.

When both implemented taxes exceed the foreign rate, the investment condition is

$$[pA(\tau^R) + (1 - p)A(\tau^O)]\rho'(K) = (1 - \tau^*)\rho'(\bar{k} - K). \quad (8)$$

The expected marginal return on host investment equals the marginal return forgone abroad. Profit shifting affects investment through the after-tax return retained under each type of government.

3.5 Tax Announcements

An equilibrium in which both types announce the same tax is a pooling equilibrium. Following [Lu \(2013\)](#), I select the commitment type's preferred common announcement, subject to both types' announcement incentive compatibility constraints. An announcement different from the prescribed rate is off path and leads firms to assign a probability of one to the opportunistic type.

Let K_0 denote investment when the government is known to be the opportunistic type. Define revenue in that equilibrium by

$$R_0 = \max_{\tau \in [0,1]} \mathcal{R}(K_0, \tau).$$

I use this known-type equilibrium after every off-path announcement. Firms then invest K_0 , and either type can collect at most R_0 on that capital. The commitment type's preferred common announcement solves

$$\begin{aligned} & \max_{\tau^R \in [0,1]} \mathcal{R}(K(\tau^R, p), \tau^R), \\ \text{subject to } & \mathcal{R}(K(\tau^R, p), \tau^R) \geq R_0, \\ & \mathcal{R}(K(\tau^R, p), \tau^O(K(\tau^R, p))) \geq R_0. \end{aligned} \tag{9}$$

The first constraint requires the commitment type to weakly prefer the common announcement over any off-path announcement. The second imposes the same requirement on the opportunistic type, which optimizes the tax after investment. [Appendix B.1](#) proves the existence of the selected pooling equilibrium and justifies the selection criterion.

I call the solution the optimal tax under commitment and denote it by $\tau^R(p)$. I compare pooling equilibria at the same prior p . The off-path belief described above applies to deviations within each equilibrium. Full commitment is the case $p = 1$ when firms know that the government will implement its announcement.

4 Reputation and Optimal Taxation

I compare optimal announced taxes across prior probabilities p , holding production technology, the foreign tax rate, and profit-shifting opportunities fixed.

In equilibrium, the opportunistic type chooses the same tax τ^O at every reputation. The tax solves

$$(2\gamma + 1)(1 - \tau^O)^3 = (1 - \tau^*)^2(1 + \tau^O). \tag{10}$$

The optimal discretionary tax exceeds the foreign tax. Firms retain a positive share $A(\tau^O) > 0$ of host profits, so investment remains positive even when $p = 0$.

Proposition 1 (Reputation lowers the optimal tax with profit shifting). *In the economy defined in Section 3, the optimal tax under commitment $\tau^R(p)$ is unique. It satisfies $\tau^* < \tau^R(p) < \tau^O$ for every $p \in (0, 1]$, and for every $0 < p < p' \leq 1$,*

$$\tau^R(p') < \tau^R(p), \quad K(\tau^R(p'), p') > K(\tau^R(p), p).$$

Proof. See Appendix B.1. □

At the announcement stage, a higher tax reduces the tax base through both investment and profit shifting. Reputation changes the investment response. Because profit shifting preserves a positive return under discretion, a marginal increase in the announced tax reduces expected returns proportionally more when reputation is higher. Higher reputation also raises investment, which reduces the elasticity of host profits with respect to expected returns. The greater sensitivity of expected returns dominates the decline in the elasticity of host profits, so the tax base becomes more elastic with respect to the announced rate. The commitment type, therefore, chooses a lower tax.

The lower announced tax and higher probability of implementation both raise investment. A lower implemented tax also reduces the share of profits shifted abroad. The commitment type collects more revenue from the larger tax base despite the lower tax rate. The opportunistic type earns more revenue at the same discretionary tax because host profits increase.

Without profit shifting, firms cannot shield host profits from taxation. The opportunistic type chooses $\tau^O = 1$. Firms receive positive after-tax host profits only when the government is the commitment type, so investment satisfies

$$p(1 - \tau^R)\rho'(K) = (1 - \tau^*)\rho'(\bar{k} - K). \quad (11)$$

At a fixed announced tax, higher reputation raises expected returns and investment. Because firms retain host profits only under the commitment type, a marginal tax increase reduces expected host returns by the same proportion at every reputation. Higher investment reduces the elasticity of host profits with respect to the after-tax return. A marginal tax increase, therefore, causes a smaller proportional contraction of the tax base, and the commitment type chooses a higher rate.

Proposition 2 (Reputation raises the optimal tax without profit shifting). *In the economy without profit shifting, under the production assumptions in Section 3, (9) has a unique maximizer $\tau^R(p) \in (0, 1)$ for every $p \in (0, 1]$. For every $0 < p < p' \leq 1$,*

$$\tau^R(p') > \tau^R(p), \quad K(\tau^R(p'), p') > K(\tau^R(p), p).$$

Proof. See Appendix B.2. □

Propositions 1 and 2 give opposite tax responses to reputation. With profit shifting, the commitment type chooses a higher announced tax when firms assign a lower probability to that type. Without profit shifting, the corresponding tax is lower. The comparison explains why the ability to move reported profits matters for the positive risk–tax relation in the data.

The announced and expected tax rates move in opposite directions without profit shifting. Although $\tau^R(p)$ rises with reputation, Proposition 2 shows that investment also rises. The investment condition (11) then requires $p(1 - \tau^R)$ to increase, so the expected tax rate $1 - p(1 - \tau^R)$ falls. The decline in the probability of confiscation therefore outweighs the increase in the announced rate.

For the numerical example, I set $z = 1$, $\alpha = 0.66$, $r^* = 0.04$, $\bar{k} = 10$, $\tau^* = 0.15$, and $\gamma = 1.25$ in $\rho(k) = zk^\alpha - r^*k$. The comparison without profit shifting uses the same production parameters. Appendix C.1 describes the numerical method.

In Figure 2, increasing p from 0.1 to 0.9 lowers τ^R from 34.3 to 30.8 percent. Investment rises, and firms shift less profit abroad when the announced tax is implemented. The resulting expansion of the reported base raises the commitment type’s revenue despite the lower rate.

The discretionary tax remains at 34.7 percent. With more capital in the host country, firms shift a larger amount of profits abroad under discretion, even though the transferred share is unchanged. Without profit shifting, the optimal tax under commitment rises from 34.4 to 42.4 percent over the same reputation range.

5 Government Reputation and Future Investment

A government that expects future tax revenue may forgo a current tax increase to preserve firms’ willingness to invest. I introduce a second period in which firms observe first-period policy before investing again.

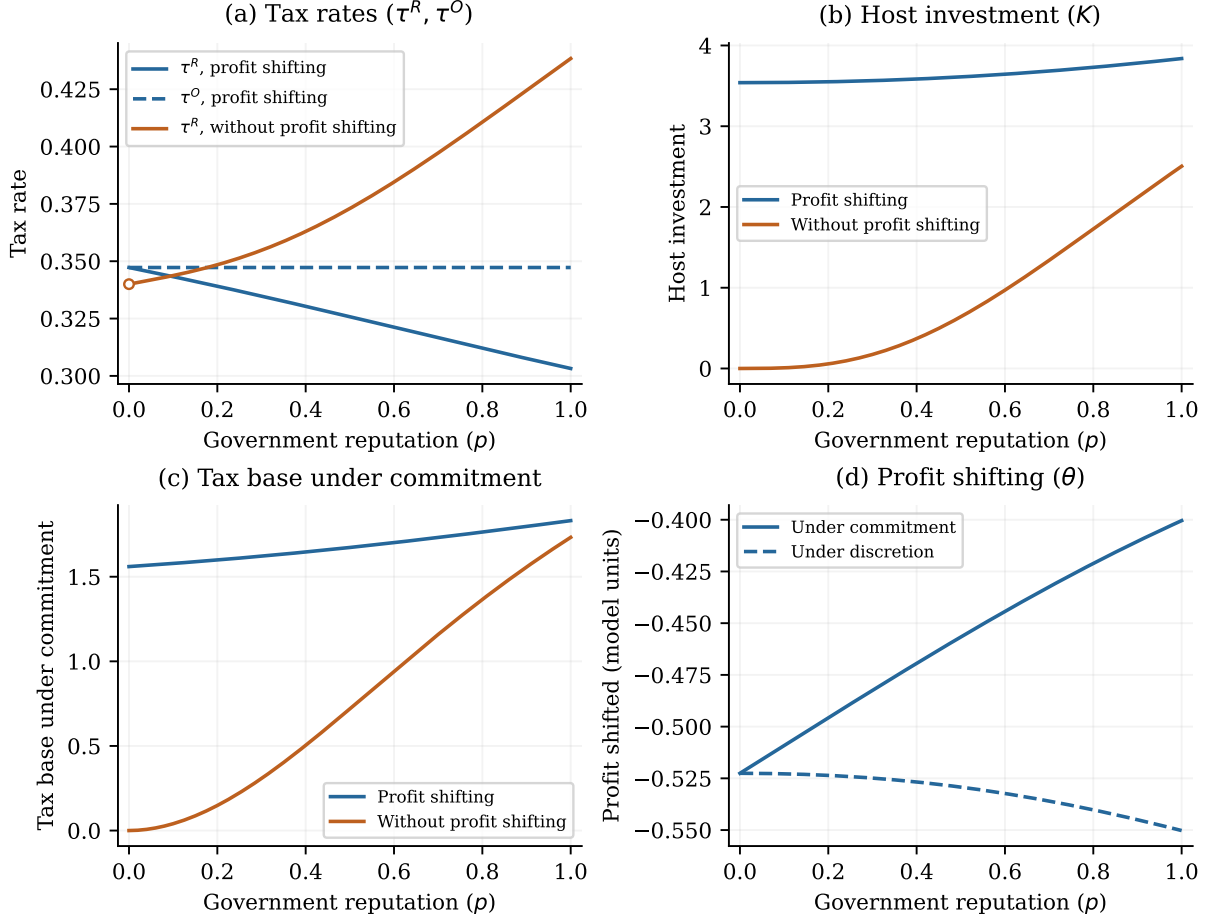


FIGURE 2: Reputation, tax policy, investment, and profit shifting

Notes: Parameters are stated in the text. Costs are incurred and deducted where profits originate. Panel (a) reports the optimal tax under commitment and the discretionary tax. The open circle denotes the limit of the announced tax as reputation approaches zero without profit shifting. Panel (c) reports the tax base under commitment. Negative transfers in panel (d) denote outward profit shifting. The discretionary tax without profit shifting is one.

5.1 The Two-Period Model

The same firms operate in both periods. Each firm rents $\bar{k}$ units of capital in each period and allocates them between the host country and the foreign location. Capital fully depreciates after production, so each period's capital stock equals that period's new investment. The rental cost r^* includes depreciation. Firms distribute their profits, and their access to capital in the next period is independent of current earnings.

Firms maximize expected discounted profits. An individual firm cannot affect aggregate capital or public beliefs, and its current capital does not survive into the next period. The firm therefore solves the one-period investment problem at each history. Government type remains fixed. The government maximizes the sum of current revenue and second-period revenue discounted by $\beta \in (0, 1)$. Initial reputation is $p \in (0, 1)$.

Within each period, the government announces a tax, firms invest, the government implements the tax, and firms choose their profit transfers. Firms observe the first-period announcement and the implemented tax before choosing second-period investment. In the second period, there is no future investment to discipline tax policy. I use the one-period equilibrium selected in Section 3.5 as the terminal equilibrium.

The technology and reporting costs are those in Section 3. With profit shifting, the discretionary tax is the constant τ^O derived in Section 4, and revenue under outward profit shifting is $\mathcal{R}(K, \tau) = v(\tau)\rho(K)$. The equilibrium at $p = 0$ gives $K_0 = K(\tau^O, 0)$ and $R_0 = \mathcal{R}(K_0, \tau^O)$. Write $V_C(q)$ and $V_O(q)$ for the two types' revenues in the selected second-period equilibrium at reputation q , with $V_C(0) = V_O(0) = R_0$.

Higher reputation in the second period induces more investment in the selected one-period equilibrium. The opportunistic type then collects revenue at the fixed rate τ^O on higher host profits, so $V_O(q)$ increases. Higher reputation also raises the commitment type's revenue at a fixed announced tax. Its maximized revenue $V_C(q)$ therefore increases as well. Both continuation revenues are also increasing in reputation without profit shifting, with $\mathcal{R}(K, \tau) = \tau\rho(K)$, $\tau^O = 1$, and $R_0 = 0$.

5.2 Tax Implementation and Tax Announcements

Let τ denote a candidate common first-period tax announcement. The commitment type implements τ . Let m be the probability that the opportunistic type also implements τ . For $\tau \neq \tau^O$, let u be the overall probability of implementation and q be the posterior probability of the commitment type after implementation. Bayes' rule gives

$$u = p + (1 - p)m, \quad q = \frac{p}{u}. \quad (12)$$

First-period investment is $K(\tau, u)$, using the investment function from the one-period model with u in place of p . Any tax revision reveals the opportunistic type and reduces reputation to zero. Conditional on a tax revision, the opportunistic type chooses τ^O because every revised tax rate yields the same continuation reputation.

The opportunistic type compares the current gain from a tax revision with the discounted loss of future revenue. Define

$$D(\tau, m; p) = \mathcal{R}(K(\tau, u), \tau^O) - \mathcal{R}(K(\tau, u), \tau) - \beta[V_O(q) - R_0]. \quad (13)$$

The opportunistic type implements the announced tax with certainty if $D \leq 0$ at $m = 1$ and revises the tax with certainty if $D \geq 0$ at $m = 0$. For $0 < m < 1$, the opportunistic type must be indifferent between implementing and revising the announced tax, so $D = 0$. Equilibrium requires the implementation incentive compatibility conditions,

firms' investment optimality, and Bayes' rule for q to hold jointly. When $m = 1$, investment is $K(\tau, 1)$ and reputation remains p . When $m = 0$, investment is $K(\tau, p)$ and observing implementation of the announced tax identifies the commitment type, giving $q = 1$.

As in Section 3.5, I select the common announcement that maximizes the commitment type's discounted revenue, subject to both types' announcement incentive compatibility constraints. Announcements above τ^O are dominated, as shown in Appendix B.3. The commitment type solves

$$\max_{\tau \in [0, \tau^O]} \{ \mathcal{R}(K(\tau, u(\tau)), \tau) + \beta V_C(q(\tau)) \}, \quad (14)$$

where $u(\tau)$ and $q(\tau)$ are determined by the implementation conditions above. An off-path announcement leads firms to assign a probability of zero to the commitment type in both periods. Either type can earn at most $(1 + \beta)R_0$ after an off-path announcement.

The commitment type's and the opportunistic type's incentive compatibility constraints at the tax announcement stage are

$$\begin{aligned} \mathcal{R}(K(\tau, u), \tau) + \beta V_C(q) &\geq (1 + \beta)R_0, \\ m[\mathcal{R}(K(\tau, u), \tau) + \beta V_O(q)] + (1 - m)[\mathcal{R}(K(\tau, u), \tau^O) + \beta R_0] &\geq (1 + \beta)R_0. \end{aligned} \quad (15)$$

Appendix B.3 establishes the existence of an optimal common announcement. The implementation probability for the opportunistic type is unique at every $\tau < \tau^O$. At $\tau = \tau^O$, both types implement the same tax rate, and reputation remains p .

An announcement with $m < 1$ raises the commitment type's continuation revenue to $V_C(q)$ because honoring the announcement increases reputation to $q = p/u > p$. The commitment type can therefore earn more from an announcement with $m < 1$ than from one implemented by both types with certainty.

5.3 Reputation, Tax Policy, and Investment

With profit shifting, a small tax cut below τ^O raises firms' after-tax returns to first order. The opportunistic type's current gain from reversing that cut is only second order because τ^O maximizes tax revenue on installed capital. The future revenue lost after a tax revision can therefore support a tax reduction that attracts additional investment. Write $\tau_1^R(p; \beta)$ for the optimal announcement in (14).

Proposition 3 (Optimal tax announcements at low reputation). *In the two-period economy with profit shifting, for every fixed $\beta \in (0, 1)$, there is $\bar{p}_\beta > 0$ such that the optimal common tax announcement is unique and strictly decreasing in p on $(0, \bar{p}_\beta)$. As reputation approaches zero, the optimal announcement converges to τ^O from below. The opportunistic type's implementation probability has a strictly positive limit, which can be below one, and investment converges to $K_0 > 0$.*

Proof. See Appendix B.3. □

Figure 3 shows how the optimal first-period tax and investment vary with reputation, with and without profit shifting. I use the same production parameters and discount factor in both cases. Let K_F denote full-commitment investment in each economy. First-period investment and host output losses are $1 - K/K_F$ and $1 - (K/K_F)^\alpha$. The benchmark differs across the two economies because profit shifting also affects investment under full commitment.

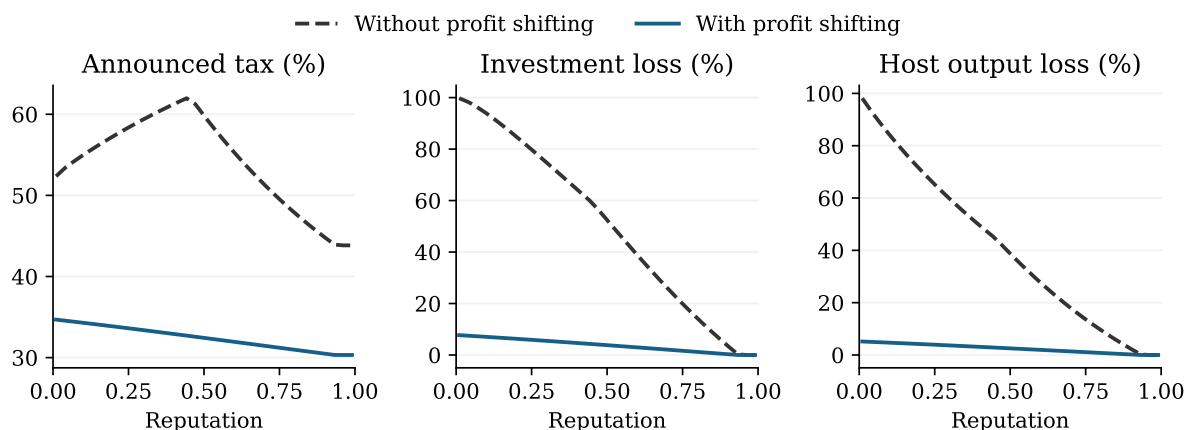


FIGURE 3: Optimal announced taxes and real activity under uncertain commitment

Notes: The common tax announcement maximizes the commitment type's discounted revenue subject to both types' incentive compatibility constraints. The discount factor is 0.6. Profit-shifting costs are incurred and deducted in the country from which profits are transferred, with $\gamma = 1.25$. Other parameters are those of Section 4. First-period investment and host output losses are measured against each economy's full-commitment allocation.

Without profit shifting, the announced tax rises from 55.0 percent at $p = 0.1$ to 61.2 percent at $p = 0.4$, then falls to 44.8 percent at $p = 0.9$. On the rising segment, the opportunistic type sometimes revises the announced tax. The probability u of facing the announced rate increases enough to raise $u(1 - \tau_1^R)$ and investment despite the higher tax. Once the opportunistic type implements the optimal announced tax with certainty, higher reputation increases the future revenue cost of a tax revision and permits a lower announced tax.

With profit shifting, both government types implement the optimal first-period announced tax at every reputation shown in Figure 3. The announced tax declines until it reaches the full-commitment rate. At $p = 0.5$, both types honor the announcement in each economy, yet first-period investment is 52.4 percent below its full-commitment level without profit shifting and 3.8 percent below with profit shifting. The corresponding host output losses are 38.8 and 2.5 percent. These losses remain because the tax needed to induce the opportunistic type to honor its announcement exceeds the full-commitment rate. The output comparison concerns production in the host, since capital not invested there remains available abroad.

Profit shifting sustains investment by preserving firms' returns after a tax revision. The opportunistic type's implementation decision depends on both the current gain from a tax revision and the future revenue loss, and profit shifting changes both amounts. In particular, firms continue to invest after learning that the government is the opportunistic type, allowing it to collect revenue despite losing its reputation.

6 Sustainable Tax Policy and Investment

Following [Chari and Kehoe \(1990\)](#), a sustainable plan specifies government policies and private decisions after every public policy history. Firms' investment and reporting choices must maximize expected discounted profits subject to the capital and tax-base constraints. Each government type must maximize discounted revenue when choosing its announcement. The commitment type must honor that announcement, while the opportunistic type must also maximize discounted revenue when choosing the implemented tax. Beliefs must satisfy Bayes' rule at histories reached with positive probability. Off-path announcements lead to reputation zero and the continuation equilibrium specified in Section 5.2.

6.1 Sustainable Taxes in Two Periods

To determine how implementation incentives limit investment, I first consider sustainable plans in which both types implement the first-period announcement. I call an announcement supported by such a plan a sustainable tax. The second period follows the selected one-period equilibrium, in which the opportunistic type may revise the announced tax. Sustainability requires the government's planned actions to be optimal at each history, including the terminal tax choice.

Under first-period implementation by both types, $u = 1$ and $q = p$. At a given tax, current investment is $K(\tau, 1)$ and does not depend on reputation. Write $H(\tau) =$

$\mathcal{R}(K(\tau, 1), \tau)$ for current revenue. The current gain from a tax revision, $G(\tau)$, and the future revenue loss, $L(p)$, enter the implementation incentive compatibility constraint

$$\underbrace{\mathcal{R}(K(\tau, 1), \tau^O) - H(\tau)}_{G(\tau)} \leq \beta \underbrace{[V_O(p) - R_0]}_{L(p)}. \quad (16)$$

Both current revenues are evaluated at the capital installed after the announcement. Firms can shift profits after a tax revision but cannot change that capital. The future loss arises because firms invest less in the second period after learning that the government is the opportunistic type.

The announcement incentive compatibility constraints are

$$\begin{aligned} H(\tau) + \beta V_C(p) &\geq (1 + \beta)R_0, \\ H(\tau) + \beta V_O(p) &\geq (1 + \beta)R_0. \end{aligned} \quad (17)$$

Continuation revenue is independent of the first-period tax when both types implement the announcement. Given the terminal equilibrium, the commitment type therefore chooses

$$\max_{\tau \in [0, 1]} H(\tau) \quad \text{subject to } G(\tau) \leq \beta L(p) \text{ and (17)}. \quad (18)$$

I denote the optimal sustainable tax by $\tau_1^{R,s}(p)$. It need not equal the optimal announcement τ_1^R in Section 5.2, which allows the opportunistic type to revise the announced tax with positive probability.

Proposition 4 (Optimal sustainable taxes with profit shifting). *In the economy with profit shifting, for any $\beta \in (0, 1)$, the optimal sustainable tax $\tau_1^{R,s}(p)$ is unique. It equals the full-commitment tax $\tau^R(1)$ if $\tau^R(1)$ satisfies (16). Otherwise, $\tau_1^{R,s}(p)$ is the unique tax in $(\tau^R(1), \tau^O)$ at which the constraint binds. For every $0 < p < p' < 1$,*

$$\tau_1^{R,s}(p') \leq \tau_1^{R,s}(p), \quad \text{strictly if } \tau_1^{R,s}(p) > \tau^R(1).$$

Both announcement incentive compatibility constraints hold at $\tau_1^{R,s}(p)$.

Proof. See Appendix B.4. □

When the full-commitment rate is infeasible, a higher announced tax reduces the current gain from a tax revision. Higher p raises second-period revenue $V_O(p)$ after the announcement is honored, while second-period revenue after a tax revision remains R_0 . The larger loss $L(p)$ therefore permits a lower tax while the implementation incentive compatibility constraint binds. The tax reduction raises current investment and revenue.

Once the full-commitment rate is sustainable, further increases in p leave current taxes and investment unchanged.

Without profit shifting, the opportunistic type can collect all current host profits by raising an announced tax below one to a rate of one. The tax revision reveals the government's type and eliminates second-period investment, creating a future revenue cost even without profit shifting.

Proposition 5 (Optimal sustainable taxes without profit shifting). *In the economy without profit shifting, under the production assumptions of Proposition 2, the optimal sustainable tax is unique. It equals $\tau^R(1)$ when the full-commitment rate is sustainable and otherwise is the unique tax above $\tau^R(1)$ satisfying*

$$(1 - \tau_1^{R,s})\rho(K(\tau_1^{R,s}, 1)) = \beta\rho(K(\tau^R(p), p)). \quad (19)$$

As reputation rises, the optimal sustainable tax weakly decreases and investment weakly increases, with strict changes while the tax exceeds $\tau^R(1)$. Both announcement incentive compatibility constraints hold.

Proof. See Appendix B.5. □

While the implementation incentive compatibility constraint binds, the optimal sustainable tax declines with reputation in both economies. Higher reputation increases the revenue lost after a tax revision and allows the government to maintain a lower tax. With a fixed tax implemented by both types, reputation leaves current investment unchanged. The investment gain comes from the lower sustainable rate. In the one-period comparison, reputation alters investment at a given announcement through the probability that the announced rate is implemented.

6.2 Sustainable Plans over a Finite Horizon

The economy now lasts T periods, indexed by $t = 1, \dots, T$, with full depreciation and the same access to capital in each period. I consider plans in which both types implement the common tax announcement at every date $t < T$. At date T , the government and firms follow the selected one-period equilibrium. Reputation remains p as long as both types honor the plan. A tax revision reveals the opportunistic type after which firms expect discretionary taxation in every subsequent period.

Given the terminal equilibrium, I determine the commitment type's preferred sustainable tax by backward induction from date $T - 1$ to date 1. At each date, the opportunistic type's future revenue determines its incentive to honor the current announcement.

Let $E_t(p)$ denote the opportunistic type's revenue from dates t through T in excess of what it would earn if its type were known before the date- t investment. Both revenues are discounted to date t . The terminal value is $E_T(p) = V_O(p) - R_0$. At each earlier date, the optimal sustainable tax $\tau_t^{R,s}(p)$ and excess revenue satisfy

$$\begin{aligned}\tau_t^{R,s}(p) &= \arg \max_{\tau} H(\tau) \quad \text{subject to } G(\tau) \leq \beta E_{t+1}(p), \\ E_t(p) &= H(\tau_t^{R,s}(p)) - R_0 + \beta E_{t+1}(p), \quad t = T-1, \dots, 1.\end{aligned}\tag{20}$$

When $T = 2$, $E_2(p) = L(p)$, and the recursion gives the two-period sustainable tax $\tau_1^{R,s}(p)$.

Proposition 6 (Sustainable taxes and investment over a finite horizon). *Under the assumptions of Proposition 4 or 5, for any fixed finite horizon $T \geq 2$, (20) selects a unique tax $\tau_t^{R,s}(p)$ at each date $t < T$. Each tax satisfies both types' announcement incentive compatibility constraints and is weakly decreasing in reputation, strictly so whenever it exceeds $\tau^R(1)$. As $p \downarrow 0$, the tax at each date $t < T$ converges to τ^O with profit shifting and to one without profit shifting. Investment, host production, and revenue at each such date converge to $K_0 > 0$, $zK_0^\alpha > 0$, and $R_0 > 0$ with profit shifting and to zero without profit shifting.*

Proof. See Appendix B.6. □

For any fixed horizon, the future revenue lost after a tax revision vanishes as the prior approaches zero. Sustainable announcements must therefore approach the discretionary rate. Firms still invest under a known opportunistic type when profit shifting preserves after-tax returns.

6.3 Investment at Low Reputation

The investment comparison extends to equilibria in which the opportunistic type may revise the announced tax before the terminal period. I now allow common announcements and implementation probabilities to depend on policy history. Government and firm decisions must remain sequentially optimal, and beliefs must satisfy Bayes' rule. Let $\mathcal{E}_T(p)$ denote this set of equilibria over T periods with initial reputation p , and let K_e denote date-1 investment in equilibrium e . The set includes all terminal common-announcement equilibria.

Proposition 7 (Investment as reputation vanishes). *For any fixed finite horizon $T \geq 1$ and $\beta \in (0, 1)$, under the production assumptions of Proposition 2, investment without profit shifting satisfies*

$$\lim_{p \downarrow 0} \sup_{e \in \mathcal{E}_T(p)} K_e = 0.$$

Date-1 host production and tax revenue consequently converge to zero. With profit shifting as defined in Section 3, there is a constant $\underline{K} > 0$ such that every equilibrium in $\mathcal{E}_T(p)$ satisfies $K_e \geq \underline{K}$, independently of p and T . The comparison does not depend on which common tax announcement the commitment type prefers.

Proof. See Appendix B.7. □

Without profit shifting, positive investment requires firms to expect that some host profits will remain untaxed. As reputation vanishes, the future revenue lost after a tax revision becomes too small to sustain investment that is bounded away from zero over a fixed finite horizon. With profit shifting, firms retain positive host returns even when the implemented tax is equal to one. The ability to report profits abroad therefore sustains investment without requiring the opportunistic type to honor its announcement.

7 Concluding Remarks

Countries with greater investment risk have higher statutory corporate tax rates. I explain this association through uncertainty about the government's commitment to taxation and firms' ability to shift profits. Profit shifting preserves returns under discretionary taxation. As reputation rises, a marginal increase in the announced tax then causes a larger proportional contraction of the tax base, inducing the commitment type to lower its rate. Without profit shifting, the tax base becomes less elastic as reputation rises and the optimal announced tax increases.

Future investment gives the opportunistic type a reason to honor its announcement. Among taxes implemented by both types, higher reputation permits a lower rate by increasing the revenue lost after a tax revision. The lower sustainable tax attracts more investment until the full-commitment allocation becomes feasible. Profit shifting also supports investment when reputation is too low to sustain these tax reductions. Over a fixed finite horizon, investment and production vanish as reputation approaches zero without profit shifting but remain positive with profit shifting, even when tax policies depend on history. Firms' ability to protect returns after investment therefore changes both optimal tax announcements and the investment consequences of limited commitment.

References

- Aguiar, M. and Amador, M. (2011). Growth in the shadow of expropriation. *The Quarterly Journal of Economics*, 126(2):651–697.
- Altshuler, R., Boller, L., Roberts, K. A., and Suárez Serrato, J. C. (2025). Tax planning and multinational behavior. Working paper, February.
- Backus, D. and Driffill, J. (1985). Rational expectations and policy credibility following a change in regime. *The Review of Economic Studies*, 52(2):211–221.
- Beer, S., de Mooij, R., and Liu, L. (2018). International corporate tax avoidance: A review of the channels, effect sizes, and blind spots. Technical Report 18/168, International Monetary Fund.
- Bilicka, K. A., Devereux, M. P., and Güçeri, İ. (2024). Tax policy, investment and profit shifting. Technical Report 33132, National Bureau of Economic Research.
- Chari, V., Kehoe, P., and Prescott, E. (1988). Time consistency and policy. Staff Report 115, Federal Reserve Bank of Minneapolis.
- Chari, V. V. and Kehoe, P. J. (1990). Sustainable plans. *Journal of Political Economy*, 98(4):783–802.
- Court of Justice of the European Union (2018). Judgment in case c-284/16, slovak republic v achmea bv. March 6. Press release 26/18.
- Devereux, M. P., Lockwood, B., and Redoano, M. (2008). Do countries compete over corporate tax rates? *Journal of Public Economics*, 92(5):1210–1235.
- Dovis, A. and Kirpalani, R. (2021). Rules without commitment: Reputation and incentives. *The Review of Economic Studies*, 88(6):2833–2856.
- European Union (2020). Agreement for the termination of bilateral investment treaties between the member states of the european union. Official Journal L 169, May 29, pp. 1–41.
- Eurostat (2026). Foreign controlled enterprises: Inward fats. Historical dataset fats_g1b_08, gross investment in tangible goods, variable V15110. Accessed September 22, 2026.
- Hong, Q. and Smart, M. (2010). In praise of tax havens: International tax planning and foreign direct investment. *European Economic Review*, 54(1):82–95.
- Hopland, A. O., Lisowsky, P., Mardan, M., and Schindler, D. (2018). Flexibility in income shifting under losses. *The Accounting Review*, 93(3):163–183.

- International Monetary Fund (2007). Haiti: Selected issues and statistical appendix. Country Report 07/292, International Monetary Fund.
- International Monetary Fund (2010). Togo: Third review under the three-year arrangement under the poverty reduction and growth facility. Country Report 10/50, International Monetary Fund.
- Johannesen, N. (2010). Imperfect tax competition for profits, asymmetric equilibrium and beneficial tax havens. *Journal of International Economics*, 81(2):253–264.
- Kallen, C. (2023). Profit shifting and multinational investment. Working paper, University of Wisconsin–Madison, January 14.
- King, R. G., Lu, Y. K., and Pastén, E. S. (2008). Managing expectations. *Journal of Money, Credit and Banking*, 40(8):1625–1666.
- Lu, Y. K. (2013). Optimal policy with credibility concerns. *Journal of Economic Theory*, 148(5):2007–2032.
- Mardan, M. and Stimmelmayer, M. (2020). Tax competition between developed, emerging, and developing countries—same same but different? *Journal of Development Economics*, 146:102491.
- McClure, C. G. (2023). How costly is tax avoidance? evidence from structural estimation. *The Accounting Review*, 98(6):353–380.
- Mongrain, S., Oh, D., and van Ypersele, T. (2023). Tax competition in the presence of profit shifting. *Journal of Public Economics*, 224:104940.
- OECD (2026). Corporate income tax rates database. Combined statutory corporate income tax rates, accessed September 22, 2026.
- Peralta, S., Wauthy, X., and van Ypersele, T. (2006). Should countries control international profit shifting? *Journal of International Economics*, 68(1):24–37.
- Phelan, C. (2006). Public trust and government betrayal. *Journal of Economic Theory*, 130(1):27–43.
- Phelan, C. and Stacchetti, E. (2001). Sequential equilibria in a ramsey tax model. *Econometrica*, 69(6):1491–1518.
- Schnitzer, M. (1999). Expropriation and control rights: A dynamic model of foreign direct investment. *International Journal of Industrial Organization*, 17(8):1113–1137.
- The PRS Group (n.d.). ICRG methodology. Methodology document, accessed September 22, 2026.

Thomas, J. and Worrall, T. (1994). Foreign direct investment and the risk of expropriation. *The Review of Economic Studies*, 61(1):81–108.

World Bank (2004). *World Development Report 2005: A Better Investment Climate for Everyone*. World Bank and Oxford University Press, Washington, DC.

A Data Appendix

A.1 Sources and Estimation Samples

The risk series comes from the Investment Profile sheet of ICRG Table 3B, January 1984–December 2021. I require twelve reported monthly scores to calculate an annual mean and subtract that mean from twelve. The three underlying subcomponent scores are not separately available in the Investment Profile sheet. Historical country aggregates are not assigned to successor countries. I merge the annual risk measure with tax rates dated one year later.

The initial estimation sample contains 1,907 country-year observations in 101 countries over 2003–2021. The main estimation sample contains 1,887 observations in the same countries. Economic controls are real GDP per capita in constant 2015 US dollars, population, and natural-resource rents/GDP from the World Bank.

The additional ICRG controls use the same twelve-month requirement and one-year lag as Investment Profile. I subtract corruption and law and order scores from six, bureaucratic quality from four, and government stability and internal and external conflict from twelve. Six, four, and twelve are the respective component maxima in the ICRG methodology. All six controls are observed throughout the tax estimation sample, so including the institutional and political controls leaves the sample unchanged.

The OECD series records zero rates for Haiti in 2003–2010 and Togo in 2003–2014. Historical IMF reports describe positive corporate tax rates during these intervals. For Haiti, the 2006 reform replaced a graduated schedule with a 30 percent proportional rate ([International Monetary Fund, 2007](#)). For Togo, IMF documentation reports positive rates before and after the 2009–2010 reductions ([International Monetary Fund, 2010](#)). I exclude twenty inconsistent zero records without imputing replacement rates, leaving 1,887 observations in 101 countries. Excluding both countries entirely leaves 1,869 observations in 99 countries. Other zero-rate jurisdictions remain in the sample.

A.2 Tax Estimates and Robustness

Standard errors are clustered by country, with the number of countries minus one as the reference degrees of freedom. Table 3 reports all slope coefficients for columns (4) and (5) of Table 1. Table 4 keeps the full set of controls fixed and compares the original OECD sample with the samples excluding the questionable zero rates and excluding Haiti and Togo entirely.

Adding lagged GDP growth gives coefficients on investment risk of 0.354, 0.420, and 0.424 for the three samples in Table 4, respectively, with p -values of 0.020, 0.005,

TABLE 3: Tax regressions with full control coefficients

	(4)	(5)
Investment risk	0.426 (0.143)	0.420 (0.145)
Log real GDP per capita	1.653 (1.337)	1.708 (1.354)
Log population	4.364 (1.894)	4.390 (1.887)
Resource rents/GDP	0.041 (0.033)	0.043 (0.033)
Corruption risk	-0.417 (0.306)	-0.427 (0.305)
Law and order risk	0.013 (0.471)	0.011 (0.473)
Bureaucracy risk	1.419 (1.029)	1.419 (1.028)
Government stability risk	0.102 (0.114)	0.098 (0.114)
Internal conflict risk	-0.336 (0.239)	-0.334 (0.240)
External conflict risk	-0.080 (0.311)	-0.088 (0.310)
Lagged GDP growth	–	-0.016 (0.024)
Country and year effects	Yes	Yes
Observations	1,887	1,887

Notes: Columns (4) and (5) of Table 1. All ICRG measures are lagged by one year and expressed as risks. Income and population are logged, and resource rents/GDP and statutory rates are in percentage points. Parentheses contain country-clustered standard errors. Dashes indicate controls omitted from a specification.

TABLE 4: Tax regressions under alternative sample restrictions

	(1)	(2)	(3)
Investment risk	0.353	0.426	0.430
Standard error	(0.150)	(0.143)	(0.143)
<i>p</i> -value	0.021	0.004	0.003
Sample	Original	Main	Exclude Haiti/Togo
Full controls	Yes	Yes	Yes
Country and year effects	Yes	Yes	Yes
Observations	1,907	1,887	1,869
Countries	101	101	99
Within R^2	0.058	0.038	0.039

Notes: All columns include economic, institutional, and political controls and country and year effects. Column (1) uses OECD rates as recorded. Column (2) uses the main sample, excluding the twenty zero-rate records in Haiti and Togo described above. Column (3) excludes both countries entirely. Parentheses contain country-clustered standard errors. No replacement rates are imputed.

and 0.004. With economic controls alone, the coefficient in the original sample is 0.263 ($p = 0.148$), compared with 0.395 ($p = 0.009$) in the main sample. In the original sample, the positive association is significant with the full set of controls but not with economic controls alone.

Reestimating the specification with all controls after omitting one country at a time gives risk coefficients between 0.280 and 0.396 in the original sample. All are positive, and 100 of the 101 estimates are significant at 5 percent. Omitting Liberia gives $p = 0.053$. After excluding the questionable zero records, the coefficients range from 0.371 to 0.457, and all estimates have $p < 0.009$. Thus no single country accounts for the positive association in the main sample.

A.3 Treaty Exposure and Investment

The investment series is V15110 in Eurostat’s inward foreign affiliates statistics. The series records gross investment in tangible goods, or annual additions to tangible assets, in millions of euros. The measure captures foreign affiliates’ expenditures rather than an FDI flow or a stock of assets. The observations are indexed by host country and the country of the ultimate controlling parent. The treatment group consists of pairs in the 2020 intra-EU termination agreement with a treaty in force immediately before the judgment in 2018. All covered treaties in the sample were in force by 2010. Both countries must fall under the agreement, and I remove pairs with a previously terminated treaty from the control group.

The estimating equation for host h , parent country j , and year t is

$$\begin{aligned} \mathbb{E}[\text{Investment}_{hjt} \mid h, j, t] = & \exp\left(\alpha_{hj} + \lambda_{ht} + \lambda_{jt} + \delta_{hj}(t - 2017) \right. \\ & \left. + \sum_s \beta_s \text{Covered treaty}_{hj} \mathbf{1}\{t = s\}\right). \end{aligned} \quad (21)$$

The indicator $\text{Covered treaty}_{hj}$ identifies covered pairs, and $\mathbf{1}\{t = s\}$ selects year s . Country-pair fixed effects absorb time-invariant differences in investment between covered and uncovered pairs, so I normalize the 2017 coefficient to zero. The country-pair trends $\delta_{hj}(t - 2017)$ in column (2) of Table 2 also absorb differences in linear investment trends, so I additionally normalize the 2010 coefficient to zero. The remaining coefficients measure departures from the fitted trends. Estimation uses Poisson pseudo-maximum likelihood.

The identifying comparison is between pairs with different treaty coverage, conditional on pair, host-year, and parent-year effects and, in column (2) of Table 2, country-pair linear trends. A causal interpretation requires the remaining investment shocks to be unrelated to coverage. Past treaty choices may be correlated with subsequent invest-

ment shocks, so the estimated coefficients need not reflect the effect of the judgment alone. An affiliate recorded under its ultimate parent's country in Eurostat may seek arbitration through an intermediate holding company with a different treaty nationality.

Standard errors are clustered by host and parent country, with 21 reference degrees of freedom in the main sample. The joint pre-event tests use host clustering because the two-way covariance matrix is not positive definite.

Reporting coverage falls sharply in 2018 and recovers in 2019 (Figure 4). Table 5 excludes 2018 and compares estimates for all reporting pairs with estimates for pairs observed in every remaining year, with and without country-pair linear trends.

The 2019 coefficients are negative in all four specifications and significant only in the balanced samples. The balanced samples also exhibit differential pre-event movements. Changes in reporting coverage and the differential pre-event movements prevent attributing the relative investment decline to the judgment alone.

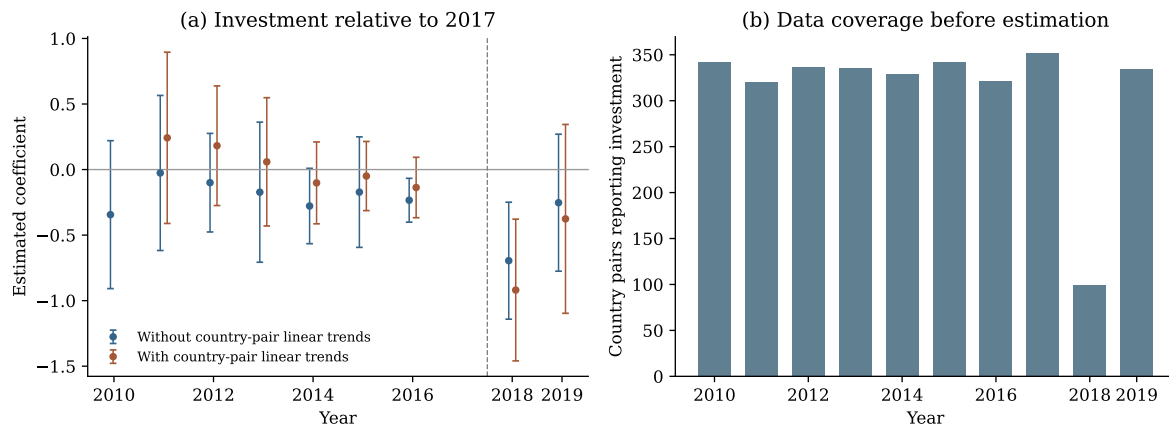


FIGURE 4: Investment around Achmea and the coverage of the data

Notes: Panel (a) reports annual coefficients and 95 percent confidence intervals for the specifications in Table 2. Both specifications include pair, host-year, and parent-year effects. The specification with pair trends uses 2010 and 2017 as reference years, and the specification without pair trends uses 2017. Standard errors are clustered by host and parent country. The vertical line marks the judgment. Panel (b) reports the number of country pairs with investment data in each year.

TABLE 5: Achmea estimates excluding the 2018 reporting gap

Country-pair linear trends	Available No	Balanced No	Available Yes	Balanced Yes
2019 treaty coefficient	-0.252	-0.523	-0.373	-0.662
Standard error	(0.250)	(0.185)	(0.348)	(0.237)
p -value	0.325	0.011	0.296	0.012
Joint pre-event p -value	0.018	< 0.001	0.546	< 0.001
Observations	2,527	1,674	2,478	1,674
Host countries	22	20	22	20

Notes: Annual event-study regressions estimated by Poisson pseudo-maximum likelihood over 2010–2019 excluding 2018, with country-pair, host-year, and parent-year effects. Balanced samples contain pairs observed in all nine remaining years. The specifications follow (21). Parentheses contain standard errors clustered by host and parent country. Joint pre-event tests use host clustering.

B Proofs

B.1 Proof of Proposition 1

Proof. Put $B = 1 - \tau^*$ and $\mu(k) = -\rho''(k)/\rho'(k)$. For $w > 0$, let $\mathcal{K}(w)$ solve

$$w\rho'(\mathcal{K}) = B\rho'(\bar{k} - \mathcal{K}).$$

The production restriction in (2) makes marginal profits positive and strictly decreasing. Their limit at zero is infinite, so $\mathcal{K}(w)$ is unique, interior, and strictly increasing. Define

$$\varepsilon(w) = \frac{d \log \rho(\mathcal{K}(w))}{d \log w}, \quad \chi(w) = -\frac{d \log \varepsilon(w)}{d \log w}. \quad (22)$$

Writing $K = \mathcal{K}(w)$, $\mu_H = \mu(K)$, and $\mu_F = \mu(\bar{k} - K)$, implicit differentiation gives

$$\varepsilon(w) = \frac{\rho'(K)}{\rho(K)(\mu_H + \mu_F)} > 0. \quad (23)$$

Its reciprocal is

$$-\frac{\rho(K)\rho''(K)}{\rho'(K)^2} + \frac{\rho(K)}{\rho'(K)}\mu(\bar{k} - K).$$

The first term increases with K when $r^* > 0$ and is constant when $r^* = 0$. The ratio $\rho(K)/\rho'(K)$ increases by concavity. Moreover,

$$-\frac{\mu'(k)}{\mu(k)^2} = \frac{1 - (2 - \alpha)r^*/(\alpha z k^{\alpha-1})}{1 - \alpha} > 0, \quad (24)$$

so $\mu(\bar{k} - K)$ also increases with K . Thus ε decreases strictly with w and $\chi > 0$. The reciprocal of ε tends to $(1 - \alpha)/\alpha$ as $K \downarrow 0$.

For the bound needed below, put $\lambda = \mu_H/(\mu_H + \mu_F)$. Log differentiation of (23) gives

$$\chi(w) = \lambda \left[1 + \frac{\rho'(K)}{\rho(K)\mu_H} \right] + \frac{\mu'(K) - \mu'(\bar{k} - K)}{(\mu_H + \mu_F)^2}.$$

Positive marginal profits imply

$$\frac{\rho'(K)}{\rho(K)\mu_H} = \frac{\alpha}{1-\alpha} \frac{[1 - r^*/(\alpha z K^{\alpha-1})]^2}{1 - r^*/(z K^{\alpha-1})} \leq \frac{\alpha}{1-\alpha}.$$

Using (24) for both countries therefore yields

$$\chi(w) \leq \frac{1}{1-\alpha} \left[1 - \lambda + (2-\alpha)\lambda^2 \frac{r^*}{\alpha z K^{\alpha-1}} \right].$$

When $K \leq \bar{k}/2$, $\mu_H \geq \mu_F$, so $1 - \lambda \leq 1/2$ and

$$\chi(w) \leq C_\rho \equiv \frac{1}{1-\alpha} \left[\frac{1}{2} + \frac{(2-\alpha)r^*}{\alpha z (\bar{k}/2)^{\alpha-1}} \right]. \quad (25)$$

Since $\varepsilon(w) = \lambda \rho'(K)/[\rho(K)\mu_H]$, the preceding inequalities also give $\varepsilon(w) \leq \alpha/(1-\alpha)$ at every investment allocation.

For outward profit shifting, write $b(\tau) = 1 - y(\tau) - \gamma y(\tau)^2/2$, so $v(\tau) = \tau b(\tau)$ and $A'(\tau) = -b(\tau)$. Before the host tax base vanishes,

$$v''(\tau) = -\frac{B^2(\tau+2)}{\gamma(1-\tau)^4} < 0. \quad (26)$$

The second restriction in (2) implies $v'(\tau^*) = 1 - \tau^*/(\gamma B) > 0$. At the tax rate where the host tax base first reaches zero, $b = 0$ and the left derivative of v is negative. Among tax rates that induce outward profit shifting, government revenue is therefore uniquely maximized at $\tau^O > \tau^*$, whose first-order condition is (10). Write $A_O = A(\tau^O)$. A transfer of share $\bar{y}$ leaves the firm $B\bar{y}$ per unit of host profit at every tax that induces outward profit shifting. The optimal transfer at τ^O is interior, so

$$A_O > B\bar{y}, \quad A(\tau^*) = B, \quad A(\tau) \geq B\bar{y} \quad \text{for } \tau \geq \tau^*. \quad (27)$$

For $\tau < \tau^*$, the inward transfer share is $x(\tau) = (\tau^* - \tau)/(B\gamma)$. The second restriction in (2) ensures $x(\tau) < \bar{y}$. The after-tax coefficients on host and foreign profits are $1 - \tau$ and

$$F(\tau) = B + (\tau^* - \tau)x(\tau) - \frac{B\gamma x(\tau)^2}{2}.$$

Since $(1 - \tau) - F(\tau) = (\tau^* - \tau)[1 - x(\tau)] + B\gamma x(\tau)^2/2 \geq 0$, the host coefficient is at least the foreign coefficient for inward transfers. Together with (27), this implies that

the ratio of host to foreign after-tax coefficients is at least $\bar{y}$ at every implemented tax. Taking expectations preserves the inequality. The investment condition therefore gives, for $K \leq \bar{k}/2$,

$$\bar{y} \leq \frac{\rho'(\bar{k} - K)}{\rho'(K)} \leq \left(\frac{K}{\bar{k} - K} \right)^{1-\alpha}, \quad \frac{\rho(K)}{\rho(\bar{k} - K)} \geq \left(\frac{K}{\bar{k} - K} \right)^\alpha.$$

Subtracting the common rental cost weakly lowers the ratio of foreign to host marginal profits and weakly raises the ratio of host to foreign profits, giving the inequalities on the right. Hence

$$\frac{\rho(K)}{\rho(\bar{k} - K)} \geq \bar{y}^{\alpha/(1-\alpha)}.$$

The same bound holds when $K > \bar{k}/2$, because the profit ratio then exceeds one. At every induced capital allocation, revenue at a tax inducing inward profit shifting satisfies

$$\frac{\partial \mathcal{R}(K, \tau)}{\partial \tau} = \rho(K) + \left[x(\tau) - \frac{\tau}{B\gamma} \right] \rho(\bar{k} - K) > 0,$$

where the last inequality follows from the third restriction in (2). Thus the same τ^O is the global discretionary optimum at every capital allocation induced by a tax announcement.

For an announcement $\tau < \tau^*$, define the expected coefficients

$$w_H = p(1 - \tau) + (1 - p)A_O, \quad w_F = pF(\tau) + (1 - p)B.$$

Investment is $\mathcal{K}(Bw_H/w_F)$. Since $F'(\tau) = -x(\tau)$,

$$-\frac{1}{1 - \tau} \leq \frac{\partial}{\partial \tau} \log \frac{w_H}{w_F} = -\frac{p}{w_H} + \frac{px(\tau)}{w_F} \leq 0.$$

The upper bound follows from $x(\tau)(1 - \tau) \leq F(\tau)$ and $x(\tau)A_O \leq B$. Using the bounds on the profit ratio and ε gives

$$\frac{1}{\rho(K)} \frac{d}{d\tau} \{ \tau [\rho(K) + x(\tau)\rho(\bar{k} - K)] \} \geq 1 - \frac{\tau^* \alpha}{(1 - \alpha)B} - \frac{\tau^*}{B\gamma} \bar{y}^{-\alpha/(1-\alpha)} > 0.$$

At τ^* , the right-hand derivatives are $b' = -1/(B\gamma)$ and $A' = -1$, so the same positive lower bound applies to the right-hand derivative of revenue. Taxes above τ^O yield lower values of both A and v than at τ^O and hence less revenue. At τ^O , $v' = 0$ and investment decreases with the announced tax for $p > 0$, so the revenue derivative is negative. Every optimal announcement is therefore interior to (τ^*, τ^O) .

For a tax in this interval, set $w = pA(\tau) + (1 - p)A_O$ and

$$f(\tau, p) = \log\{v(\tau)\rho(\mathcal{K}(w))\}.$$

Here $w \leq B$, so $\mathcal{K}(w) \leq \bar{k}/2$ and (25) applies. Differentiation gives

$$\begin{aligned} f_\tau &= \frac{v'}{v} + A' \frac{p\varepsilon(w)}{w}, \\ f_{\tau p} &= \frac{A'\varepsilon(w)}{w^2} \{A_O - p\chi(w)[A(\tau) - A_O]\}. \end{aligned} \quad (28)$$

The fourth restriction in (2) is equivalent to $C_\rho(1/\bar{y} - 1) < 1$. Since $A(\tau) \leq B$ and $p \leq 1$,

$$A_O - p\chi(w)[A(\tau) - A_O] \geq A_O - C_\rho(B - A_O) > B[(1 + C_\rho)\bar{y} - C_\rho] > 0.$$

Thus $f_{\tau p} < 0$.

To establish uniqueness, use $v = -\tau A'$ and

$$A''(\tau) = \frac{B^2}{\gamma(1 - \tau)^3} > 0, \quad A'''(\tau) = \frac{3B^2}{\gamma(1 - \tau)^4} > 0.$$

At a stationary point, $v'/v = -pA'\varepsilon/w$. Substitution into the second derivative gives

$$f_{\tau\tau} = \frac{A'' + \tau A''' - \tau(A'')^2/A'}{\tau A'} - \left(\frac{v'}{v}\right)^2 \left[1 + \frac{1 + \chi(w)}{\varepsilon(w)}\right] < 0. \quad (29)$$

Every stationary point is a strict maximum. Two maxima would require a stationary minimum between them. The derivative signs at τ^* and τ^O therefore establish a unique global optimum. The implicit-function theorem and (28) give

$$\frac{d\tau^R(p)}{dp} = -\frac{f_{\tau p}}{f_{\tau\tau}} < 0.$$

Continuity extends the strict tax ordering to $p' = 1$.

The tax-base elasticity satisfies

$$-\frac{\partial}{\partial \log \tau} \log \left[\frac{\mathcal{R}(K(\tau, p), \tau)}{\tau} \right] = 1 - \tau f_\tau(\tau, p).$$

Its derivative with respect to reputation is $-\tau f_{\tau p} > 0$ at each fixed tax in (τ^*, τ^O) . At the optimum, $f_\tau = 0$, so the tax-base elasticity equals one.

Both a higher prior and a lower announced tax increase w and investment. At each fixed tax in (τ^*, τ^O) , higher reputation also raises the commitment type's revenue, so

maximized revenue increases strictly. At $p = 0$, $K_0 = \mathcal{K}(A_O)$ and $R_0 = v(\tau^O)\rho(K_0)$. The commitment type can obtain R_0 by announcing τ^O and strictly improves on it at its interior optimum. The opportunistic type earns $v(\tau^O)$ per unit of host profit. Investment exceeds K_0 and increases with reputation, so its revenue exceeds R_0 and rises strictly. Both announcement incentive compatibility constraints hold. Finally, the announcement and its outward transfer share are below their discretionary counterparts and decline with reputation. The expected tax rate and expected outward transfer share consequently decline as well. $\square$

The proof also establishes the known-type equilibrium. At $p = 0$, the unique investment K_0 and the discretionary tax τ^O are mutual best responses. After the selected common announcement, firms retain prior p and choose $K(\tau^R(p), p)$. The commitment type implements the announcement, and the opportunistic type imposes τ^O . After an off-path announcement, firms assign probability one to the opportunistic type and invest K_0 . Either type can then collect at most R_0 . The announcement constraints verified above rule out profitable deviations. Firms optimize reporting after every history, so these strategies and beliefs form a weak perfect Bayesian equilibrium.

To verify the announcement selection, first consider separation. An opportunistic type identified before investment earns R_0 . If the commitment type announces $\tau < \tau^O$, firms invest more than K_0 , and the opportunistic type gains by imitating that announcement before imposing τ^O . If the commitment type announces $\tau > \tau^O$, it earns less than R_0 and can gain by announcing τ^O . Any surviving separating candidate therefore gives both types R_0 . Both strictly prefer the selected common announcement.

Under the strongly coherent criterion of [Lu \(2013\)](#), an off-path announcement belonging to an alternative pooling equilibrium that strictly benefits the commitment type must induce a posterior at least as high as p . The commitment type chooses that announcement with probability one, while the opportunistic type does so with probability at most one. At any common announcement that does not maximize the commitment type's revenue, moving to its maximizing announcement gives a strict gain. Revenue at that announcement increases with the posterior, so the deviation remains profitable under the required belief. A separating candidate is ruled out in the same way by a common announcement near the optimum and outside the separating path, which benefits both types. Conversely, no alternative pure tax-announcement equilibrium improves the commitment type's payoff at the selected optimum. Only the opportunistic type can prefer an alternative equilibrium, so the off-path belief assigning probability zero to the commitment type satisfies the criterion and supports the selected allocation.

B.2 Proof of Proposition 2

Proof. For $p > 0$ and $\tau < 1$, investment without profit shifting satisfies (11), so $K = K(p(1 - \tau))$, $K_\tau < 0$, and $K_p > 0$. The tax-base elasticity is

$$-\frac{\partial \log \rho(K)}{\partial \log \tau} = \frac{\tau}{1 - \tau} \varepsilon(p(1 - \tau)).$$

It decreases with reputation because ε decreases strictly. At an interior optimum, the elasticity equals one. Equivalently,

$$\tau^R(p) = 1 - \left\{ 1 + \left[-\frac{\rho''(K)}{\rho'(K)} - \frac{\rho''(\bar{k} - K)}{\rho'(\bar{k} - K)} \right] \frac{\rho(K)}{\rho'(K)} \right\}^{-1}. \quad (30)$$

The expression in braces increases strictly with K , as shown in the proof of Proposition 1. Thus the tax rate satisfying the first-order condition increases with capital. Substituting (30) into

$$p = \frac{B}{1 - \tau^R} \frac{\rho'(\bar{k} - K)}{\rho'(K)}$$

gives a strictly increasing function of K . This function tends to zero as $K \downarrow 0$ and to infinity as $K \uparrow \bar{k}$. For every $p > 0$, there is therefore exactly one tax rate satisfying the first-order condition, and both the tax and capital increase with p . Revenue is positive at interior taxes and zero at zero and one, so this rate is the global maximizer. Higher capital also requires a larger expected retained share $p(1 - \tau^R)$. The expected tax rate $1 - p(1 - \tau^R)$ consequently decreases with reputation.

A known opportunistic type imposes a tax of one, firms invest no capital in the host, and revenue is zero. Each type earns positive revenue at the selected announcement, so both announcement incentive compatibility constraints hold. The equilibrium and selection argument for Proposition 1 applies with $\tau^O = 1$ and $K_0 = R_0 = 0$. In particular, a separating commitment-type announcement below one can be profitably imitated by the opportunistic type, while an announcement of one yields zero revenue. At any nonmaximizing common announcement, the commitment type gains by moving to its maximizing rate, and revenue at that rate increases with the posterior. The same off-path belief therefore supports the selected pooling equilibrium and satisfies Lu's criterion. $\square$

B.3 Proof of Proposition 3

Proof. Fix $p \in (0, 1)$ and $\beta \in (0, 1)$, with the terminal equilibrium and off-path beliefs in Section 5.2. For $\tau < \tau^O$, investment increases with the implementation probability m . Under outward profit shifting, the current gain from a tax revision is $[v(\tau^O) -$

$v(\tau)]\rho(K(\tau, u))$, which increases with investment. Under inward profit shifting, let $x(\tau)$ be the transfer per unit of foreign profit. The gain is

$$[v(\tau^O) - \tau]\rho(K(\tau, u)) - \tau x(\tau)\rho(\bar{k} - K(\tau, u)).$$

This also increases with investment because $v(\tau^O) > \tau^* > \tau$ and foreign profits fall as host capital rises. Without profit shifting, the gain is $(1 - \tau)\rho(K(\tau, u))$ and has the same property. Meanwhile, $q = p/[p + (1 - p)m]$ falls with m , reducing $V_O(q)$. Hence $D(\tau, m; p)$ is strictly increasing in m for $\tau < \tau^O$.

The implementation probability is therefore uniquely determined by the endpoint inequalities or the interior equation $D = 0$. It is continuous in τ . At $\tau = \tau^O$, both types implement the same tax and reputation remains p . Since $V_O(p) > R_0$, both types also implement announcements sufficiently close to τ^O from below. The allocation is thus continuous on $[0, \tau^O]$. The rate τ^O satisfies both announcement constraints, so their feasible set is nonempty and closed.

Announcements above τ^O are dominated. At such an announcement, investment is at most K_0 , current revenue under commitment is below R_0 , and discretionary revenue is at most R_0 . When $m = 1$, announcing τ^O gives higher current revenue and the same continuation. If $m < 1$, the current gain from a tax revision is at least $\beta[V_O(q) - R_0]$, with equality when $m > 0$. Holding u fixed, the gain from a tax revision falls continuously from $v(\tau^O)\rho(K(0, u))$ to zero as the announcement rises from zero to τ^O . The gain at zero is at least the gain at the original announcement. There is therefore a $\tilde{\tau} < \tau^O$ at which the gain equals $\beta[V_O(q) - R_0]$. The same m and q satisfy the implementation conditions at $\tilde{\tau}$. Investment is now above K_0 , and the commitment type receives discretionary revenue on that larger capital stock less $\beta[V_O(q) - R_0]$. Its current revenue strictly increases, its continuation is unchanged, and the opportunistic type's payoff weakly increases. Both announcement constraints remain satisfied. Continuity on the compact feasible set in $[0, \tau^O]$ then establishes existence of an optimal common announcement.

For the low-reputation result, consider the economy with profit shifting. Set $B = 1 - \tau^*$ and let $\mathcal{K}(w)$ solve $w\rho'(\mathcal{K}) = B\rho'(\bar{k} - \mathcal{K})$. Write $S(w) = \rho(\mathcal{K}(w))$, $A_O = A(\tau^O)$, and $K_0 = \mathcal{K}(A_O)$. Equation (26) gives $v''(\tau^O) < 0$. Since $A'(\tau^O) < 0$ and $S'(A_O) > 0$, define

$$\kappa_R = \frac{v(\tau^O)A'(\tau^O)S'(A_O)}{v''(\tau^O)\rho(K_0)} > 0.$$

The terminal announcement satisfies

$$v'(\tau^R)S(w) + v(\tau^R)S'(w)qA'(\tau^R) = 0, \quad w = A_O + q[A(\tau^R) - A_O].$$

At $(\tau^R, q) = (\tau^O, 0)$, the left-hand side has derivative $v''(\tau^O)\rho(K_0) \neq 0$ with respect to τ^R . The implicit-function theorem gives $\tau^R(q) = \tau^O - \kappa_R q + O(q^2)$. Thus, for $L(q) = V_O(q) - R_0$ and $J(q) = V_O(q) - V_C(q)$,

$$L(q) = \ell q^2 + O(q^3), \quad J(q) = \frac{1}{2}\ell q^2 + O(q^3), \quad \ell = [-v''(\tau^O)]\rho(K_0)\kappa_R^2 > 0. \quad (31)$$

Indeed, $w - A_O = [-A'(\tau^O)]\kappa_R q^2 + O(q^3)$ and $v(\tau^O) - v(\tau^R) = [-v''(\tau^O)]\kappa_R^2 q^2/2 + O(q^3)$. Proposition 1 also gives $J(q) > 0$ for every $q > 0$, since $\tau^R(q) < \tau^O$.

Suppose an optimizing sequence as $p \downarrow 0$ had $q \geq \delta > 0$. Then $u = p/q \rightarrow 0$. The expected host and foreign return coefficients converge uniformly to A_O and B , so investment converges to K_0 , including for announcements below τ^* . When $0 < m < 1$, the binding implementation constraint gives the commitment type the value

$$v(\tau^O)\rho(K) + \beta R_0 - \beta J(q). \quad (32)$$

With $m = 0$, its value is at most this amount with $q = 1$. Continuity and positivity of J on $[\delta, 1]$ therefore put the limiting payoff strictly below $(1 + \beta)R_0$. Announcing τ^O instead yields $R_0 + \beta V_C(p) \rightarrow (1 + \beta)R_0$. When $m = 1$, $q = p \rightarrow 0$ directly. Every optimizing sequence must therefore have $q \rightarrow 0$ and, for sufficiently small p , $m > 0$.

The reporting formulas imply a positive lower bound on investment at every announced tax. A tax bounded away from τ^O would consequently leave a positive gain from a tax revision while $L(q) \rightarrow 0$, violating implementation incentives when $m > 0$. Hence every optimal announcement approaches τ^O . By the preceding dominance argument, it induces outward profit shifting at sufficiently small p .

Parameterize the binding implementation constraint by q , with $u = p/q$ and $q \geq p$,

$$[v(\tau^O) - v(\tau)]S \left(A_O + \frac{p}{q}[A(\tau) - A_O] \right) = \beta L(q). \quad (33)$$

Substitute $\tau = \tau^O - qZ$ and divide by q^2 . The equation extends smoothly to the origin, where it becomes

$$\frac{[-v''(\tau^O)]\rho(K_0)Z^2}{2} = \beta \ell.$$

Its derivative with respect to Z is nonzero at the positive solution. The implicit-function theorem gives a smooth $Z(p, q)$ with $Z(0, 0) = \kappa_R \sqrt{2\beta}$. At $q = p$, both types implement $\tau = \tau^O - pZ(p, p)$, and the implementation constraint binds. For sufficiently small p , this tax exceeds $\tau^R(1)$. Current revenue decreases with the tax above $\tau^R(1)$, as established in Appendix B.1, so this rate maximizes the commitment type's revenue among announcements implemented by both types.

Along the binding constraint, the commitment type's value is

$$U(p, q) = v(\tau^O)S \left(A_O + \frac{p}{q} [A(\tau^O - qZ(p, q)) - A_O] \right) - \beta J(q) + \beta R_0.$$

Its smooth extension satisfies

$$\begin{aligned} U(0, 0) &= (1 + \beta)R_0, & U_q(0, 0) &= 0, \\ U_p(0, 0) &= v(\tau^O)S'(A_O)[-A'(\tau^O)]\kappa_R\sqrt{2\beta} > 0, \\ U_{qq}(0, 0) &= -\beta\ell < 0. \end{aligned}$$

Thus U is strictly concave in q near the origin. Expanding its first-order condition gives

$$U_q(p, q) = U_{pq}(0, 0)p - \beta\ell q + O((|p| + |q|)^2).$$

The implicit-function theorem gives a unique unconstrained maximizer

$$\hat{q}(p) = \lambda_\beta p + O(p^2), \quad \lambda_\beta = \frac{U_{pq}(0, 0)}{\beta\ell}.$$

The constrained optimum is $q(p) = \max\{p, \hat{q}(p)\}$. The preceding convergence argument rules out other optimal announcements at sufficiently small p .

At this optimum, the expected retained share satisfies

$$w - A_O = [-A'(\tau^O)]\kappa_R\sqrt{2\beta}p + O(p^2), \quad J(q) = O(p^2).$$

The opportunistic type's value is $v(\tau^O)S(w) + \beta R_0$, and the commitment type's value is lower by $\beta J(q)$. Both exceed $(1 + \beta)R_0$ by

$$v(\tau^O)S'(A_O)[-A'(\tau^O)]\kappa_R\sqrt{2\beta}p + O(p^2) > 0$$

for sufficiently small p . Both announcement constraints therefore hold.

Substituting $q(p)$ into the tax and implementation probability yields

$$\begin{aligned} \tau_1^R(p; \beta) &= \tau^O - c_\beta p + O(p^2), \\ c_\beta &= \kappa_R\sqrt{2\beta}\max\{1, \lambda_\beta\} > 0, \quad \lim_{p \downarrow 0} m(p) = \frac{1}{\max\{1, \lambda_\beta\}} > 0. \end{aligned} \quad (34)$$

Each local tax branch has derivative converging to $-c_\beta$ when it is selected. If $\lambda_\beta = 1$, both branches have the same strictly negative limiting derivative. The selected tax is therefore strictly decreasing at sufficiently small p and approaches τ^O from below. Strict concavity in q and uniqueness of the commitment type's preferred tax among

announcements implemented by both types imply that the optimal announcement is unique. Since $w \rightarrow A_O$, investment converges to $K_0 > 0$. $\square$

B.4 Proof of Proposition 4

Proof. Write $\tau^F = \tau^R(1)$. Proposition 1 and its proof establish a unique τ^O , a unique $\tau^F \in (\tau^*, \tau^O)$, and continuation revenues that increase strictly with reputation. When both types implement a tax in $[\tau^*, \tau^O]$,

$$H(\tau) = v(\tau)\rho(K(\tau, 1)), \quad G(\tau) = [v(\tau^O) - v(\tau)]\rho(K(\tau, 1)).$$

Investment decreases with the tax. Appendix B.1 shows that H increases up to τ^F and decreases thereafter. Since $v' > 0$ on $[\tau^*, \tau^O]$,

$$G'(\tau) = -v'(\tau)\rho(K(\tau, 1)) + [v(\tau^O) - v(\tau)]\rho'(K(\tau, 1))K_\tau(\tau, 1) < 0.$$

Below τ^* , investment decreases and current revenue increases with the tax by the inward-transfer calculation in Appendix B.1. Discretionary revenue therefore decreases, so G decreases there as well. Thus G decreases continuously to zero at τ^O . Above τ^O , current revenue is below R_0 because both investment and revenue per unit of host profit are lower.

Fix $p \in (0, 1)$. Both continuation revenues are at least R_0 , and $L(p) > 0$. Any current revenue of at least R_0 therefore satisfies both announcement constraints. The tax τ^O is feasible and yields R_0 , whereas τ^F yields strictly more. If $G(\tau^F) \leq \beta L(p)$, the unconstrained maximizer τ^F is feasible and uniquely optimal. Otherwise, strict monotonicity of G gives a unique $\tau \in (\tau^F, \tau^O)$ satisfying $G(\tau) = \beta L(p)$. Every lower tax violates implementation incentives. Every higher tax up to τ^O yields less revenue, and taxes above τ^O yield less than R_0 . This tax rate is therefore uniquely optimal and satisfies both announcement constraints.

Higher reputation raises $L(p)$ and lowers the optimal sustainable tax while the implementation constraint binds, increasing investment and current revenue. Once τ^F is feasible, further increases in reputation leave current taxes and investment unchanged. $\square$

B.5 Proof of Proposition 5

Proof. Without profit shifting, $\tau^O = 1$, $R_0 = 0$, and

$$H(\tau) = \tau\rho(K(\tau, 1)), \quad G(\tau) = (1 - \tau)\rho(K(\tau, 1)).$$

Investment is positive and strictly decreasing for $\tau < 1$. Proposition 2 and its proof give a unique revenue maximum at $\tau^F = \tau^R(1)$, with H increasing below τ^F and decreasing above it. Both positive factors in G decrease with the tax, so G is strictly decreasing and converges to zero as $\tau \uparrow 1$.

The opportunistic type loses future revenue $L(p) = \rho(K(\tau^R(p), p))$ after a tax revision. This loss is positive and strictly increasing by Proposition 2. The argument for Proposition 4 applies. The optimum is τ^F when $G(\tau^F) \leq \beta L(p)$ and otherwise the unique tax above τ^F satisfying $G(\tau) = \beta L(p)$. While this constraint binds, the optimal sustainable tax decreases and investment increases with reputation. Both types earn positive revenue, while an off-path announcement yields zero, so both announcement constraints hold. $\square$

B.6 Proof of Proposition 6

Proof. Let $E \geq 0$ denote the opportunistic type's continuation revenue in excess of what it earns when its type is known. The preceding two proofs give a unique optimal sustainable tax $\tau(E)$. It equals $\tau^F = \tau^R(1)$ if $G(\tau^F) \leq \beta E$ and otherwise solves $G(\tau(E)) = \beta E$ on $(\tau^F, \tau^O]$. Thus $\tau(E)$ is continuous and nonincreasing, strictly decreasing when it exceeds τ^F . Current revenue $H(\tau(E))$ is continuous, nondecreasing in E , and at least R_0 .

The map $E \mapsto H(\tau(E)) - R_0 + \beta E$ is continuous, strictly increasing, and zero at $E = 0$. The unique static optimum and the maximum theorem imply continuity of the terminal value $E_T(p) = V_O(p) - R_0$. The investment condition gives $E_T(p) \rightarrow 0$ as $p \downarrow 0$, and Propositions 1 and 2 give strict monotonicity. Applying (20) backward from date T shows that each $E_t(p)$ is continuous, strictly increasing, and converges to zero as $p \downarrow 0$. At every date $t < T$, $\tau_t^{R,s}(p) = \tau(E_{t+1}(p))$ therefore decreases with reputation whenever the implementation constraint binds above τ^F and converges to τ^O as $p \downarrow 0$. Investment converges to $K_0 > 0$ with profit shifting and to zero without profit shifting, where $\tau^O = 1$. Continuity of production and current revenue gives the remaining limits.

Following an off-path announcement at date t , firms invest at reputation zero, and either type can earn at most $R_0 \sum_{j=0}^{T-t} \beta^j$. Revenue at each prescribed tax before date T and both terminal revenues are at least R_0 , so both types' announcement constraints hold at every date. The recursion imposes the implementation constraint. After a tax revision or an off-path announcement, discretion and firms' investment choices at reputation zero are sequentially optimal. Firms also optimize at every prescribed history. These choices and the stated beliefs support the selected plan. $\square$

B.7 Proof of Proposition 7

Proof. Without profit shifting, let $\mathcal{K}(w)$ solve $w\rho'(\mathcal{K}) = B\rho'(\bar{k} - \mathcal{K})$, where $B = 1 - \tau^*$, and set $S(w) = \rho(\mathcal{K}(w))$. Both functions are continuous and strictly increasing on $[0, 1]$, with $\mathcal{K}(0) = S(0) = 0$. Current revenue is at most $S(1)$, so discounted revenue over T periods is bounded by $M_T = S(1) \sum_{j=0}^{T-1} \beta^j$. Once the opportunistic type is revealed, backward induction gives confiscation and zero investment at every subsequent date.

Let $B_T(\delta)$ be the supremum of the opportunistic type's discounted revenue over all sequentially rational plans lasting T periods, with common announcements and initial reputation in $[0, \delta]$. Policies may depend on history, and future reputation need not remain below δ . I prove by induction on T that $B_T(\delta) \rightarrow 0$ as $\delta \downarrow 0$ and that date-1 investment converges uniformly to zero over these plans.

For $T = 1$, a tax of one is strictly optimal for the opportunistic type whenever capital is positive. Thus any equilibrium with positive investment and an announcement $\tau < 1$ has expected untaxed share $w = p(1 - \tau) \leq p$, giving $B_1(\delta) \leq S(\delta)$. An announcement of one gives zero investment directly.

Suppose the claims hold for plans lasting $T - 1$ periods. At a date-1 announcement $\tau < 1$, write $u = p + (1 - p)m$, $q = p/u$, and $w = u(1 - \tau)$. If investment did not converge uniformly to zero, some sequence with $p \downarrow 0$ would have $w \geq \epsilon > 0$. Then $u \geq \epsilon$ and $1 - \tau \geq \epsilon$, so $m > 0$ at sufficiently small p . The opportunistic type must weakly prefer implementing the announced tax to confiscating profits, implying

$$\epsilon S(\epsilon) \leq (1 - \tau)S(w) \leq \beta B_{T-1}(p/\epsilon),$$

because $q \leq p/\epsilon$. The right side converges to zero, a contradiction. An announcement of one gives zero current investment. Current investment and revenue therefore vanish uniformly.

At a given history, let $W_{T-1}(q)$ be the opportunistic type's discounted continuation revenue over the remaining $T - 1$ periods after honoring the date-1 announcement. To bound its expected continuation revenue, fix $\eta \in (0, 1)$. If $q \leq \eta$, this value is at most $B_{T-1}(\eta)$. If $q > \eta$, Bayes' rule gives $u = p/q < p/\eta$ and $m \leq u/(1 - p)$. Hence

$$mW_{T-1}(q) \leq B_{T-1}(\eta) + \frac{p}{\eta(1 - p)} M_{T-1}.$$

Continuation revenue after a tax revision is zero. Taking $p \downarrow 0$ and then $\eta \downarrow 0$ proves that expected continuation revenue also vanishes uniformly. At $\tau = 1$, the implemented tax conveys no information and discounted continuation revenue is at most $\beta B_{T-1}(p)$. This completes the induction. Production and current revenue vanish because $zK^\alpha \rightarrow 0$ and $0 \leq \mathcal{R}(K, \tau) \leq \rho(K)$.

With profit shifting, the proof of Proposition 1 establishes that the host after-tax return coefficient is at least $\bar{y}$ times the foreign coefficient at every implemented tax. Taking expectations preserves this inequality. The investment condition gives

$$K \geq \underline{K} > 0, \quad \bar{y}\rho'(\underline{K}) = \rho'(\bar{k} - \underline{K}).$$

This lower bound is independent of reputation, implementation probabilities, and horizon length. Host production is at least $z\underline{K}^\alpha > 0$. The bounds in both cases hold across all equilibria with common announcements. $\square$

C Numerical Methods

C.1 Numerical Solution

For the baseline model, the discretionary tax solves (10). For each candidate announcement, I solve the firm's investment problem and calculate the commitment type's revenue. I then choose the revenue-maximizing announcement subject to both types' announcement incentive compatibility constraints. The maximization includes boundary solutions and tax rates at which the direction of profit shifting changes or a tax-base constraint binds.

For the two-period model, I solve the implementation conditions in Section 5.2 at each candidate announcement and maximize the commitment type's discounted revenue subject to both announcement constraints. The maximization includes tax rates at which the opportunistic type's implementation probability reaches zero or one. For sustainable taxes, I maximize current revenue subject to (16) and (17). The finite-horizon calculation applies (20) backward from the terminal equilibrium.

C.2 Alternative Profit-Shifting Costs

The reporting costs in the model concern transfers within an existing multinational structure. Bilicka et al. (2024) and Altshuler et al. (2025) distinguish these costs from the costs of establishing that structure. McClure (2023) discusses operational costs, audit exposure, and financial-reporting incentives. I compare the baseline with costs incurred in both countries, with and without deductibility.

When costs are incurred and deducted in both countries, the reporting problem is

$$\begin{aligned} \max_{\theta} \quad & (1 - \tau) \left[\rho(k) + \theta - \frac{\gamma\theta^2}{2\rho(k)} \right] \\ & + (1 - \tau^*) \left[\rho(\bar{k} - k) - \theta - \frac{\gamma\theta^2}{2\rho(\bar{k} - k)} \right], \end{aligned} \quad (35)$$

subject to nonnegative tax bases in both countries. When the constraints are slack, the optimal transfer is

$$\theta(k, \tau) = \frac{(\tau^* - \tau)\rho(k)\rho(\bar{k} - k)}{\gamma[(1 - \tau)\rho(\bar{k} - k) + (1 - \tau^*)\rho(k)]}. \quad (36)$$

The feasible interval is

$$\begin{aligned} \theta &\geq \max \left\{ \frac{\rho(k)(1 - \sqrt{1 + 2\gamma})}{\gamma}, \frac{\rho(\bar{k} - k)(-1 - \sqrt{1 + 2\gamma})}{\gamma} \right\}, \\ \theta &\leq \min \left\{ \frac{\rho(k)(1 + \sqrt{1 + 2\gamma})}{\gamma}, \frac{\rho(\bar{k} - k)(-1 + \sqrt{1 + 2\gamma})}{\gamma} \right\}. \end{aligned} \quad (37)$$

The objective is strictly concave in the transfer. The constrained optimum therefore equals the transfer in (36) when it is feasible and the nearest endpoint of the feasible interval otherwise. If operating profits are zero in either country, only a zero transfer is feasible.

With non-deductible costs in both countries, the reporting objective is

$$\begin{aligned} & (1 - \tau)[\rho(k) + \theta] + (1 - \tau^*)[\rho(\bar{k} - k) - \theta] \\ & - \frac{\gamma\theta^2}{2} \left[\frac{1}{\rho(k)} + \frac{1}{\rho(\bar{k} - k)} \right]. \end{aligned} \quad (38)$$

The unconstrained transfer is $(\tau^* - \tau)\rho(k)\rho(\bar{k} - k) / [\gamma(\rho(k) + \rho(\bar{k} - k))]$. The nonnegative tax bases restrict the transfer to $[-\rho(k), \rho(\bar{k} - k)]$. For both alternatives, I solve investment and discretionary taxation jointly and compare the commitment type's revenue across common announcements satisfying both types' announcement incentive compatibility constraints.

I choose γ in each cost specification to match the tax-base semi-elasticity at equal tax rates and equal operating profits, holding capital fixed. This semi-elasticity is the negative derivative of the log host tax base with respect to the host tax. It equals $1/[\gamma(1 - \tau^*)]$ under the baseline cost, $1/[2\gamma(1 - \tau^*)]$ when costs are deducted in both countries, and $1/(2\gamma)$ when costs in both countries are not deductible. The reference semi-elasticities are 1.26 from [Kallen \(2023\)](#) and 1.50 from [Beer et al. \(2018\)](#). The latter

TABLE 6: Optimal taxes with alternative profit shifting costs

Costs incurred in	Tax deductible?	Tax-base semi-elasticity	Tax under commitment (%)		
			$p = 0.1$	$p = 0.5$	$p = 0.9$
Both countries	Yes	1.26	31.5	29.3	27.2
Both countries	No	1.26	38.1	33.8	30.2
Country from which profits are shifted	Yes	1.26	30.6	29.3	27.9
Both countries	Yes	1.50	29.2	27.3	25.6
Both countries	No	1.50	34.3	30.9	28.0
Country from which profits are shifted	Yes	1.50	28.5	27.3	26.2

Notes: Taxes are in percent. The reference semi-elasticities are evaluated at equal tax rates and equal operating profits, holding capital fixed. Production parameters are the same across cost specifications. Tax bases must be nonnegative.

also equals the short-run aggregate estimate in Bilicka et al. (2024), which holds capital and tax-planning capacity fixed.

Table 6 reports the optimal one-period taxes for each cost specification. At both reference semi-elasticities, the optimal announced tax decreases across the three reported reputation levels. Under the baseline cost specification, Table 7 reports first-period investment and host output losses relative to full commitment in the two-period model.

TABLE 7: First-period real activity relative to full commitment

Tax-base semi-elasticity	Investment loss (%)			Host output loss (%)		
	$p = 0.1$	$p = 0.5$	$p = 0.9$	$p = 0.1$	$p = 0.5$	$p = 0.9$
Without profit shifting	93.91	52.44	3.51	84.22	38.77	2.33
Baseline	7.05	3.82	0.30	4.71	2.54	0.20
1.26	4.96	2.68	0.20	3.30	1.77	0.13
1.50	4.05	2.18	0.16	2.69	1.44	0.11

Notes: Allocations under the optimal common first-period tax announcement with $\beta = 0.6$. Losses are relative to each economy's full-commitment allocation. The baseline cost specification has $\gamma = 1.25$ and tax-base semi-elasticity $1/[\gamma(1 - \tau^*)] = 0.941$ at equal tax rates and equal operating profits. For the rows labeled 1.26 and 1.50, γ is chosen to match the stated semi-elasticity. Production parameters are unchanged. In every reported allocation with profit shifting, the opportunistic type honors the first-period announcement with certainty. Without profit shifting, it sometimes revises the announced tax at $p = 0.1$ and honors it with certainty at $p = 0.5$ and 0.9 .